What the occupation functional of a finite Markov chain determines: principal minors, modal supports and cycle trees

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Abstract

What does the occupation measure of a finite Markov chain determine about its generator? For an irreducible generator Q on d states, the Donsker and Varadhan functional of the occupation measure determines exactly the principal minors of $-Q$. Kim and Landim (Stochastic Process. Appl. 189 (2025) 104733) showed that it determines a reversible generator, and their 3-cycle shows that it does not determine a nonreversible one. By known matrix theory the generators with these minors form a finite set. It is Q and its time reversal when $-Q$ has no rank-one cut, and in general it comes from an orbit of cut-transposes. For a chain that switches about $\tau \log N$ times in N steps, the first-order support of the most likely vector of counts under killing at known rates determines exactly the same minors. For an Arrhenius family with all rates positive, with long runs first and low temperature second, the functionals determine the depth of every face and the Freidlin and Wentzell cycle tree, without reversibility or nondegeneracy. A set is a cycle exactly when every proper subset is strictly shallower. A design fixed in advance reads the same tree. Under a field of size $c\,e^{-\gamma/\eta}$ on a face, the largest exponent γ at which the face still carries the mode is at most its depth, with equality for cycles, and the depth of a face is the largest such threshold over its subfaces.

Keywords: identifiability, principal minors, occupation measure, large deviations, Perron root, metastability, Freidlin–Wentzell cycles
2020 MSC: 60J27, 60F10, 60J10, 15A15, 15A18, 15A29, 15B48

1. Introduction

Suppose that we record how long a continuous-time Markov chain on $[d] = \{1, \dots, d\}$ spends in each state, and not the order of its visits or its jumps. What does this record determine about the generator Q? We compare 3 resolutions of the law of the occupation measure. The first is the law itself. The second is its logarithmic scale, the Donsker and Varadhan functional J_V [12], which is the rate function of the occupation measure. The third is a finite label, the support of the most likely vector of counts of a discretized chain killed at known rates. This paper answers the question for the second and the third, and for their low-temperature limits.

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Kim and Landim [20, Section 5] began this inverse problem for J_V. Their Lemma 5.1 reads the exit rates q_i and the edge products $q_{ij}q_{ji}$ off J_V, that is, the principal minors of orders 1 and 2 of $-Q$. Together with the stationary law, which J_V gives through its zero set (their Lemma 5.2), this fixes Q when Q is reversible (their Corollary 5.3). Their Example 5.4 shows that the uniform 3-cycle and its reversal have the same J_V, so J_V does not determine a nonreversible generator, and they call their conditions optimal. By their Corollary 5.8 the joint functional of the empirical measure and the empirical flow determines the rates when every state is recurrent. What J_V determines for a nonreversible chain was left open.

1.1. The Donsker and Varadhan functional

Theorem A (what the Donsker and Varadhan functional determines)**.** *Let Q and Q' be irreducible generators on $[d]$. Then $J_V = J'_V$ on the simplex if and only if $-Q$ and $-Q'$ have the same principal minors. In that case the exit rate $\lambda_F(v)$ of every face F under every field v, and the functional J_F of the chain killed on leaving F, are the same for Q and Q'.*

This is Theorem 3.3. It has 3 inputs. The first is the variational formula $\lambda_V(v) = \min_\alpha(J_V(\alpha) - v \cdot \alpha)$ for $\lambda_V(v) = -\varrho(Q + \operatorname{diag} v)$. The second is the irreducibility of the multiaffine polynomial $\det(-Q_F - \operatorname{diag} w)$ for every irreducible face, which is Lemma 5.1 of Al Ahmadieh [1]. The third is the Perron-branch lemma (Lemma 3.2). It says that the Perron root on an open set of fields determines that polynomial, hence every principal minor. The 3-cycle of Kim and Landim is the case $d = 3$ of the next statement.

The set of generators with given principal minors comes from the matrix literature. Write $Q^* = D_\pi^{-1} Q^\top D_\pi$ for the time reversal, with π the stationary law. A partition $[d] = X \cup Y$ with $|X|, |Y| \geq 2$ is a cut of Q if the blocks $Q[X, Y]$ and $Q[Y, X]$ have rank at most 1. Across a cut, a cut-transpose of $-Q$ replaces the 2 rank-one blocks by the products of their factors taken crosswise and transposes the block on Y (Section 3.3). The cut-transpose orbit of $-Q$ is the set of matrices reached from $-Q$ or from $-Q^\top$ by finitely many cut-transposes, each across a cut of the current matrix, up to positive diagonal similarity. Each of them has exactly one generator point, the generator $-D^{-1}CD$ obtained from a member C by a positive diagonal similarity (Definition 3.5).

Corollary 1.1 (the generator fiber)**.** *Let Q be an irreducible generator. The generators with the principal minors of $-Q$ are the generator points of the cut-transpose orbit of $-Q$. They form a finite set of irreducible generators with the spectrum of Q, which contains Q and Q^*. It is $\{Q, Q^*\}$ when Q has no cut, in particular when $d \leq 3$, and $\{Q\}$ when Q is reversible. In general it can contain generators other than Q and Q^*.*

This is Corollary 3.6 with the example after it. The fiber of the principal-minor map of an irreducible matrix, up to diagonal similarity and transposition, is due to Loewy [23], Hartfiel and Loewy [18], Al Ahmadieh [1] and Chatterjee, Ghosh, Gurjar and Raj [7]. Our part is a short translation to generators. A Perron and Frobenius normalization picks exactly one generator in each class of diagonal similarity, and a cut-transpose acts as Q on one side of the cut and as Q^* on the other (Proposition 3.9).

The fiber is not an artefact of the logarithmic scale. From the stationary start the law of the counts is the same under Q and Q^*, by path reversal (Remark 3.8). For a start outside a finite union of proper subspaces, the constants of the local law separate every 2 members of the fiber (Proposition 3.9).

1.2. Modal supports

The label enters through a chain that rarely switches. Fix a start law and run the chain for N steps with transition matrix $I + (T/N)Q$, so that it makes about T switches. Its composition is the vector of occupation counts, and the support of a composition is a face of the simplex. For $T = \tau \log N$ the most likely composition on a face F has probability $N^{-\Phi_\tau(F)+o(1)}$, with the score $\Phi_\tau(F) = |F| - 1 + \tau\lambda_F$ and λ_F the exit rate of F [25]. One power of N per dimension competes with the cost of staying in the face. So the face of the mode is a nontrivial function of τ, and it becomes a function of (τ, v) when each composition is weighted by $e^{Tv\cdot n/N}$ for a field v. For $v \le 0$ this weight is the chance that the chain survives when it is killed at state i at the known rate $-v_i$. The first-order modal-support map W_Q records which face minimizes the score with the field, on the open dense set of (τ, v) where exactly one face does (Section 4.1).

Theorem B (what the first-order modal supports determine)**.** *Let Q and Q' be irreducible generators and let the start have full support. Then $W_Q = W_{Q'}$ outside a closed nowhere dense set of (τ, v) if and only if $-Q$ and $-Q'$ have the same principal minors. If the minors differ, there is a nonempty open set of (τ, v) on which both maps have a strict winner and the 2 winners differ.*

This is Theorem 4.6. The proof is a descent through the faces. For every irreducible face F and a generic field on F, the exposure lemma (Lemma 4.5) finds a proper subface G that shares a boundary of strict regions with F. Along that boundary $\lambda_F = \lambda_G - (|F| - |G|)/\tau$. Induction on $|F|$ traces λ_F on an open set of fields, and the Perron-branch lemma extends it to all fields.

The modal support is an artificial observable. The survival probability under calibrated killing is finer, it is easier to sample, and it gives every principal minor at once (Remark 3.4). We know of no experiment in which the modal support is observed and the survival probability is not. We study the label because the forward theory of [25] describes the law of the counts through it. The word *exactly* applies to W_Q only. The finite-horizon label, the support of the most likely composition at a given N, determines W_Q and so at least the minors, and whether it determines more is open (Section 6).

1.3. Low temperature and the cycle tree

Let the rates be $q_{ij} = e^{-V_{ij}/\eta}$ with fixed costs $V_{ij} \ge 0$ for all $i \ne j$, so that every rate is positive, and let the time scale be $\tau = e^{h/\eta}$ for a height h. We take $N \to \infty$ first at fixed η and then $\eta \downarrow 0$, and the observer knows η. The depth $D(F)$ of a proper face is the exponential order of its exit rate, and it is the largest depth of a Freidlin and Wentzell cycle inside F (Theorem 5.1). Away from finitely many heights the winner is a cycle of least size among those deeper than h (Theorem 5.2). At a critical height a noncycle can win on an open window of the rescaled time scale (Theorem 5.4 and Example 5.5).

So winners need not be cycles. By Theorems A and B at each η, the functionals J_V^η, or the maps W_{Q_η}, determine every exit rate, hence every depth. But they do so through the whole functional, or through fields chosen after Q_η is known. We also want a design fixed in advance. Readings at one fixed field fail. The unperturbed path shows the depth only of the faces it selects. Killing the complement of a face at one large rate keeps the face

winning above its depth, and by the shift rule rewarding the face at the same rate is the same experiment. What works is to scan the size of the field on the calibrated scale and read a threshold. For a proper face G, a constant $c > 0$ and a real γ put $v^{G,\gamma} = c\, e^{-\gamma/\eta} \mathbf{1}_G$, equivalently a killing at rate $c\, e^{-\gamma/\eta}$ off G. A face G wins at (h, v) if for all small η it is the unique minimizer of the score $|F| - 1 + e^{h/\eta} \lambda_F(v)$, so that it carries the mode for all large N at each such η. The threshold $\vartheta(G)$ is the supremum of the γ for which G wins under $v^{G,\gamma}$ at some height $h > \gamma$.

Theorem C (the cycle tree)**.** *Let (Q_η) be an Arrhenius family with all rates positive, and let the start have full support.*

(i) The functionals J_V^η, along any sequence $\eta \to 0$, determine the depth of every face, the cycles, their depths and the cycle tree. So do the maps W_{Q_η}.

(ii) For every proper face G, $\vartheta(G) \le D(G)$, with equality when G is a cycle, and $D(F) = \max_{G \subseteq F} \vartheta(G)$ for every proper face F.

(iii) A nonempty proper set C is a cycle if and only if $D(C') < D(C)$ for every nonempty proper subset C' of C.

No step uses reversibility, and no nondegeneracy of the depths is assumed.

Part (i) is Corollary 5.13, part (ii) is Theorem 5.11 and part (iii) is Theorem 5.12. Part (ii) is a design fixed before Q is known. It needs the temperature, the time unit, the amplitude c and the labels of the states, and no rate of Q, and with (iii) it reads the cycle tree from finitely many thresholds. Criterion (iii) holds with ties between the depths of different faces, because cycles are strict and depths increase strictly along nested cycles (Lemma 5.10). Its weak form, with $D(C') \le D(C)$, fails already on 3 states. Finally the endpoints of the critical windows determine the prefactors of the faces on the lower envelope and only bound the others (Theorem 5.14).

Comparison with Γ-expansions. Landim [21] for nonreversible chains and Bertini, Gabrielli and Landim [5] for reversible ones expand the level-two rate function of a family of finite chains along the metastable time scales. Kim and Landim [20] do the same for the joint functional of the measure and the flow. Their families have rates with a limit that is positive on some edge, and every ratio of products of rates is comparable [20, (2.5)]. An Arrhenius family satisfies this after a rescaling of time, since every such ratio is $e^{-b/\eta}$ for some real b. So we do not claim a weaker hypothesis. Their expansions give the metastable states and wells on each time scale, and the reduced chains between them. Theorem C adds the depth of every face, not only of the metastable wells, the criterion (iii) that recognizes a cycle from depths alone, and a readout from finitely many calibrated observations. Criterion (iii) is in neither Section 4 of Catoni [6] nor Sections 2 and 3 of Cirillo, Nardi and Sohier [8], which characterize cycles by communication heights and state the weak monotonicity of depth. We did not check Trouvé [27], and we claim no more.

A design that reads a reversible generator from $m + d - 1$ boundaries of W_Q, its stability and its finite-horizon form are in the final section of the arXiv version of this paper.

1.4. Related work

The rate inside a face is the Donsker and Varadhan functional of the killed chain [12], and the exact law of the counts goes back to Good [17]. The joint rate of the empirical measure and the flow is that of Bertini, Faggionato and Gabrielli [3, 4]. The matrix theory behind the fiber is credited above and in Section 3.3, together with Engel and Schneider [13]. Cohen [10] and Friedland [16] study the convexity of $v \mapsto \varrho(A + \operatorname{diag} v)$, and Delmas, Dronnier and Zitt [11] prove a multiplicative analogue of the Perron-branch lemma. The cycle framework is that of Freidlin and Wentzell [15], Catoni [6], Olivieri and Vares [24], Trouvé [27] and Cirillo, Nardi and Sohier [8], and the eigenvalue asymptotics behind the prefactors go back to Wentzell [28].

Identifiability from coarse observations has a long history for aggregated Markov chains, where one observes the lumped path, that is, which class of states the chain occupies at each time, as in models of ion channels. Fredkin and Rice [14], Rydén [26], Larget [22] and Bruno, Yang and Pearson [9] study when two generators give the same law of the lumped path, and give canonical forms for the equivalence classes. Xiang, Zhou, Deng and Yang [29] identify the generator of a stationary chain, reversible or in a class of irreversible ones, from hitting-time distributions observed at some states. Our data are of another kind. The occupation measure keeps every state and forgets the order of the visits.

1.5. Plan of the paper

Section 2 fixes the notation and states the results of [25] that we use. Section 3 proves Theorem A and Corollary 1.1. Section 4 proves Theorem B, and Section 5 treats the low temperature and Theorem C. Section 6 lists the computations that check the paper and the open questions.

2. Setting and imported results

We fix the notation and state, without proof, the results of [25] that we use, each with a pointer to its place there.

2.1. Setting

The chain. Fix $d \geq 2$ and a generator Q on $[d] = \{1, \dots, d\}$. Its rates are $q_{ij} \geq 0$ for $i \neq j$, the exit rates are $q_i = \sum_{j \neq i} q_{ij}$, $Q_{ii} = -q_i$, and $q_{\max} = \max_i q_i$. We write q for the vector (q_i). The start μ is a probability law on $[d]$, and $V = [d]$ is the face of all states. For $N \geq 2$ and $T > 0$ put $L = \log N$ and $\tau = T/L$. The chain $X_1, \dots, X_N$ has $X_1 \sim \mu$ and transition matrix $I + (T/N)Q$, which is stochastic when $Tq_{\max} \leq N$. The composition is the vector n of visit counts, $n_i = \#\{t \leq N : X_t = i\}$, so $n_1 + \cdots + n_d = N$. We write $P_N(n)$ for its law, $\operatorname{supp} n = \{i : n_i > 0\}$ and $\alpha = n/N$ for the visit fractions.

Faces. A face is a nonempty set $F \subseteq [d]$. Let Δ_F be the set of probability vectors on $[d]$ that vanish off F, and Δ_F° the set of those that are positive on F. The killed generator of F is $Q_F = (Q_{ij})_{i,j \in F}$. Its diagonal keeps the full exit rates q_i, so it generates the chain killed when it leaves F. For $A \subseteq F$ the principal submatrix of $-Q_F$ on A is $-Q_A$. The switching graph G_F has an edge $i \to j$ whenever $i, j \in F$, $i \neq j$ and $q_{ij} > 0$. We call F irreducible if G_F is strongly connected, so every singleton is irreducible. We call F admissible if some

path of positive probability has support exactly F, and we say that F satisfies (H) if it is irreducible and $\mu(F) > 0$.

Perron roots and fields. A square matrix B is Metzler if its off-diagonal entries are nonnegative, and its Perron root $\varrho(B)$ is then its largest real eigenvalue. If B is irreducible, $\varrho(B)$ is simple and has positive left and right eigenvectors [2]. For a face F and a field $v \in \mathbb{R}^d$ put

$$\lambda_F(v) = -\varrho\big(Q_F + \operatorname{diag} v_F\big), \qquad \Phi_{\tau,v}(F) = |F| - 1 + \tau\lambda_F(v),$$

and $\lambda_F = \lambda_F(0)$, $\Phi_\tau = \Phi_{\tau,0}$. The number λ_F is the exit rate of F, and $\lambda_{\{i\}}(v) = q_i - v_i$. If F is reducible, Q_F is block triangular in an order of its strongly connected components, and $\lambda_F(v)$ is the least of the numbers $\lambda_C(v)$ over these components C. The shift rule $\lambda_F(v + c\mathbf{1}) = \lambda_F(v) - c$ holds for every face. If F is irreducible, let r and l be positive right and left Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$, and put $\alpha_F^*(v) = l \circ r$, the quasi-ergodic law of F under the field, where $x \circ y$ is the entrywise product. Finally

$$M_F(T, v) = \max\big\{P_N(n)\, e^{Tv\cdot n/N} : \operatorname{supp} n = F\big\}, \qquad \max \emptyset = 0.$$

The functional. For $\alpha \in \Delta_F$ let

$$J_F(\alpha) = \sup_{u \in (0,\infty)^F} \sum_{i \in F} \alpha_i \Big(-\frac{(Q_F u)_i}{u_i}\Big),$$

the Donsker and Varadhan functional of the chain killed on leaving F [12], and J_V that of the whole chain.

2.2. Imported results

Lemma 2.1 (admissible faces, [25, Lemma 2.2])**.** (a) *An irreducible face is admissible if and only if $\mu(F) > 0$. So* (H) *holds exactly for the irreducible admissible faces, and if μ has full support every irreducible face is admissible.* (b) *If Q has symmetric support, that is $q_{ij} > 0$ exactly when $q_{ji} > 0$, then every admissible face is irreducible.* (c) *A face that is not admissible carries no composition of positive probability.*

Here (a) and (b) are [25, Lemma 2.2(b) and (c)], and (c) is immediate from the definition, as noted in [25, Theorem 4.2(ii)].

Lemma 2.2 (monotonicity in the face, [25, Lemma 3.2])**.** *If $F \subsetneq G$ are faces and G is irreducible, then $\lambda_G(v) < \lambda_F(v)$ for every field v. In particular every proper face of an irreducible chain has $\lambda_F > \lambda_V = 0$.*

Proposition 2.3 (the variational formula, [25, Propositions 3.1 and 3.4])**.** *Let F be a face.*

(a) $J_F = J_V$ on Δ_F.

(b) If F is irreducible, then for every $v \in \mathbb{R}^F$

$$\lambda_F(v) = \min_{\alpha \in \Delta_F} \big(J_F(\alpha) - v \cdot \alpha\big),$$

with the minimum attained only at $\alpha_F^*(v)$. So $-\lambda_F$ is the Legendre transform of J_F, and J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°. The map $v \mapsto \lambda_F(v)$ is concave and real-analytic, with $\partial\lambda_F/\partial v_i = -\alpha_F^*(v)_i$.

(c) If F is irreducible, Q_F is reversible with respect to some $\nu \in (0,\infty)^F$ and $S_F = D_{\sqrt{\nu}}(-Q_F)D_{\sqrt{\nu}}^{-1}$, then $\lambda_F(v) = \lambda_{\min}(S_F - \operatorname{diag} v_F)$.

The derivative in (b) is the first-order perturbation $\partial\varrho/\partial v_i = l_i r_i$ of a simple eigenvalue, with $l \cdot r = 1$.

Lemma 2.4 (a uniform upper bound, [25, Lemma 4.1 and (6)])**.** *Let F be irreducible with $k = |F| \geq 2$ and let $\mathcal{V} \subset \mathbb{R}^d$ be compact. There is C, depending on Q, μ and $\mathcal{V}$, such that for all $v \in \mathcal{V}$, all $N \geq 2$ and $T > 0$ with $Tq_{\max} \leq N/2$, and every n with $\operatorname{supp} n = F$,*

$$P_N(n)\, e^{Tv\cdot n/N} \leq C\Big(\frac{T}{N}\Big)^{k-1} e^{-T\lambda_F(v)+CT^2/N}. \tag{1}$$

The bound comes from the Perron tilt $\theta = q - v - \lambda_F(v)\mathbf{1}$, and it holds for a path of any length in place of N, hence for each block of a path.

For a singleton the only composition is Ne_i, so by [25, (1)]

$$M_{\{i\}}(T,v) = \mu_i\Big(1 - \frac{Tq_i}{N}\Big)^{N-1} e^{Tv_i} = \mu_i e^{-T(q_i - v_i)}\big(1 + O(T^2/N)\big). \tag{2}$$

Theorem 2.5 (first-order selection with a field, [25, Theorem 4.2 and Remark 4.3])**.** *Fix Q, μ and a field v.*

(a) Let F be admissible with $k = |F|$. There are κ, T_0 and N_0 such that $\log M_F(T,v) \leq -(k-1)L + (k-1)\log T - T\lambda_F(v) + \kappa$ for $N \geq N_0$ and $T_0 \leq T \leq N^{1/3}$. If F is not admissible, then $M_F(T,v) = 0$.

(b) Let $T = \tau L$ with $\tau > 0$ fixed, and let $v = 0$ or let μ have full support. Then

$$\log \max_n P_N(n) e^{Tv\cdot n/N} = -L \min_F \Phi_{\tau,v}(F) + O(\log L)$$

over the admissible faces, and for large N the support of every maximizer minimizes $\Phi_{\tau,v}$ among them.

At $v = 0$, part (a) is the upper bound of [25, Theorem 4.2(ii)] and (b) is [25, Theorem 4.2(iii)]. The field forms of both are [25, Remark 4.3], which derives them from the same proofs with $Q_F + \operatorname{diag} v_F$ in place of Q_F. We use (b) only at a fixed τ.

Definition 2.6 (tilt and constant, [25, Definition 5.1])**.** Let F be irreducible with $k = |F| \geq 2$, with its states numbered $1, \dots, k$. For $\alpha \in \Delta_F^\circ$ the supremum defining $J_F(\alpha)$ is attained at a vector r_α, unique up to a factor. Put $l_\alpha = \alpha/r_\alpha$ entrywise, and let $\widehat{Q}_\alpha$ be the generator on F with off-diagonal entries $q_{ij}r_{\alpha,j}/r_{\alpha,i}$, whose stationary law is α. With $D = \operatorname{diag}\alpha$ let $Z_\alpha = (\mathbf{1}\alpha^\top - \widehat{Q}_\alpha)^{-1} - \mathbf{1}\alpha^\top$ and $\Gamma_\alpha = DZ_\alpha + Z_\alpha^\top D$, and let Σ_α be Γ_α with the row and column of state 1 deleted. Then

$$C_F(\alpha) = \frac{(\mu\cdot r_\alpha)(l_\alpha\cdot\mathbf{1})}{(2\pi)^{(k-1)/2}\sqrt{\det\Sigma_\alpha}}, \qquad K_F(v) = C_F\big(\alpha_F^*(v)\big),$$

and $K_{\{i\}}(v) = \mu_i$.

Proposition 2.7 (the constant, [25, Propositions 3.3 and 5.2]). *Let F be irreducible with $k \geq 2$. The maps $\alpha \mapsto r_\alpha$ (normalized), l_α, Σ_α, $J_F(\alpha)$ and $C_F(\alpha)$ are real-analytic on Δ_F°, Σ_α is positive definite, and $\nabla^2 J_F(\alpha) = \Sigma_\alpha^{-1}$ in the coordinates $(\alpha_2, \dots, \alpha_k)$. At $\alpha = \alpha_F^*(v)$ the vector r_α is a multiple of the right Perron vector of $Q_F + \operatorname{diag} v_F$, and l_α is the matching multiple of the left one.*

The first sentence is [25, Proposition 5.2(a)]. For the second let r and l be the right and left Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$, so that $\alpha_F^*(v) = l \circ r$. The vector $u = r$ gives $-(Q_F r)_i / r_i = v_i + \lambda_F(v)$, so its value in the supremum defining $J_F(\alpha_F^*(v))$ is $\lambda_F(v) + v \cdot \alpha_F^*(v)$, which is $J_F(\alpha_F^*(v))$ by [25, Proposition 3.3(a)]. So r attains the supremum, and by the uniqueness in [25, Proposition 3.3(b)] it is a multiple of r_α. Then $l_\alpha = \alpha / r_\alpha$ is the matching multiple of l. At $v = 0$ this is the text after [25, Definition 5.1].

Theorem 2.8 (face maxima with a field, [25, Theorem 5.7]). *Let F satisfy* (H) *with $k = |F|$, and let $\mathcal{V} \subset \mathbb{R}^d$ be compact. Uniformly for $v \in \mathcal{V}$ and $1 \leq T \leq N^{1/3}$,*

$$\log M_F(T, v) = -(k-1)L + \tfrac{k-1}{2} \log T + \log K_F(v) - T\lambda_F(v) + O_{\mathcal{V}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big), \tag{3}$$

and for $k \geq 2$ every maximizer $\hat{n}$ satisfies $\hat{n}/N = \alpha_F^*(v) + O_{\mathcal{V}}(1/T)$.

Theorem 2.9 (the stratified local law, [25, Theorem 5.6]). *For every face F satisfying* (H) *with $k = |F| \geq 2$ and every compact $\mathcal{K} \subset \Delta_F^\circ$, uniformly for $\operatorname{supp} n = F$, $n/N \in \mathcal{K}$ and $1 \leq T \leq N^{1/3}$,*

$$-\log P_N(n) = (k-1)L + T\, J_V(n/N) - \frac{k-1}{2} \log T - \log C_F(n/N) + O_{\mathcal{K}}\Big(\frac{1}{T} + \frac{T^2}{N}\Big).$$

By Theorem 2.9, J_V is the logarithmic scale of the law on single compositions. Nothing else from [25] enters.

3. The Donsker and Varadhan functional and the principal minors

This section proves Theorem A. In Section 3.1 the Perron root of a face on an open set of fields determines the principal minors of the face. Section 3.2 deduces that J_V determines exactly the principal minors of $-Q$. Section 3.3 describes the generators with given minors, and Section 3.4 what the exact law adds.

3.1. The polynomial of a face and the Perron branch

The principal minors of $-Q_F$ are the numbers $\det(-Q_A)$ for $A \subseteq F$, with $\det(-Q_\emptyset) = 1$. They are the coefficients of the multiaffine polynomial

$$R_F(w) = \det\big(-Q_F - \operatorname{diag} w\big) = \sum_{A \subseteq F} \det(-Q_A) \prod_{i \in F \setminus A} (-w_i), \qquad w \in \mathbb{R}^F.$$

For every field v, $\lambda_F(v)$ is a real root of $t \mapsto R_F(v + t\mathbf{1})$, and if F is irreducible it is the smallest real root, since the other real eigenvalues of $-Q_F - \operatorname{diag} v_F$ exceed the Perron eigenvalue.

Lemma 3.1 (irreducibility)**.** *If F is irreducible, then R_F is irreducible over $\mathbb{C}$. If F is reducible, R_F is the product of the polynomials of the strongly connected components of G_F.*

Proof. Al Ahmadieh [1, Lemma 5.1] proves, over any field and for every order, that a square matrix A is irreducible exactly when $\det(\operatorname{diag} x + A)$ is an irreducible polynomial in x. Take the field $\mathbb{C}$ and $A = -Q_F$. Then $R_F(w) = \det(\operatorname{diag}(-w) + A)$, and the linear change of variables $x = -w$ preserves irreducibility. For a reducible face, $-Q_F$ is block triangular with the blocks of the components on the diagonal, and the determinant factors accordingly. □

The proof of Al Ahmadieh rests on a theorem of Hartfiel and Loewy [18, Theorem 1], that every entry of the adjugate of $\operatorname{diag} x + A$ is a nonzero polynomial when A is irreducible.

Lemma 3.2 (the Perron branch)**.** *Let F be irreducible, let Q' be another generator whose face F is irreducible, and let $c \in \mathbb{R}$. Suppose that $\lambda'_F(v) = \lambda_F(v) + c$ for all v in a nonempty open set $\mathcal{U} \subseteq \mathbb{R}^F$. Then $-Q'_F$ and $-Q_F + cI$ have the same principal minors, and $\lambda'_F = \lambda_F + c$ on all of $\mathbb{R}^F$. In particular, for $c = 0$, $-Q'_F$ and $-Q_F$ have the same principal minors.*

Proof. Let $k = |F|$ and number the states of F as $1, \dots, k$. Put $\widetilde{R}(w) = \det(-Q_F + cI - \operatorname{diag} w) = R_F(w - c\mathbf{1})$, which is irreducible over $\mathbb{C}$ by Lemma 3.1. For $k = 1$ the hypothesis says $q'_1 = q_1 + c$, which is the claim. Let $k \geq 2$, and put $\psi(v) = v + \lambda'_F(v)\mathbf{1}$. For $v \in \mathcal{U}$ the number $\lambda'_F(v) = \lambda_F(v) + c$ is an eigenvalue of $-Q'_F - \operatorname{diag} v$ and of $-Q_F + cI - \operatorname{diag} v$, so $R'_F(\psi(v)) = 0 = \widetilde{R}(\psi(v))$.

Both polynomials are affine in w_1. Write $\widetilde{R} = p_1 w_1 + p_0$ and $R'_F = p'_1 w_1 + p'_0$ with p_1, p_0, p'_1, p'_0 polynomials in $w' = (w_2, \dots, w_k)$. The coefficient of $\prod_{i \geq 2}(-w_i)$ in p_1 is -1, so $p_1 \neq 0$. The polynomial $\mathcal{D} = p_1 R'_F - p'_1 \widetilde{R} = p_1 p'_0 - p'_1 p_0$ does not involve w_1 and vanishes on $\psi(\mathcal{U})$. We claim that the projection of $\psi(\mathcal{U})$ to w' contains a nonempty open set. Fix $v^0 \in \mathcal{U}$ and consider $y \mapsto (\psi(v^0 + (0, y)))_{i \geq 2}$ for small $y \in \mathbb{R}^{k-1}$. By Proposition 2.3(b), λ'_F is real-analytic with $\partial \lambda'_F / \partial v_i = -\beta_i$, where $\beta = \alpha'^*_F(v) > 0$ and $\sum_i \beta_i = 1$. The derivative of the map at $y = 0$ is $I - \mathbf{1}\beta'^{\top}$ with $\beta' = (\beta_i)_{i \geq 2}$, of determinant $\beta_1 > 0$, and the inverse function theorem proves the claim. So $\mathcal{D} = 0$ and $p_1 R'_F = p'_1 \widetilde{R}$. The irreducible $\widetilde{R}$ divides $p_1 R'_F$, and it involves w_1, so it does not divide $p_1 \neq 0$, which is free of w_1. Hence it divides R'_F. Both have total degree k and the coefficient 1 at $\prod_i(-w_i)$, so $R'_F = \widetilde{R}$. Finally $\lambda'_F(v)$ is the smallest real root of $t \mapsto R'_F(v + t\mathbf{1}) = \widetilde{R}(v + t\mathbf{1})$, which is $\lambda_F(v) + c$. □

Delmas, Dronnier and Zitt [11, Theorem 2.1] prove a multiplicative analogue for $\eta \mapsto \varrho(K \operatorname{diag} \eta)$ on $\{0,1\}^n$, by an induction over the cube. Cohen [10] proved that $v \mapsto \varrho(A + \operatorname{diag} v)$ is convex for a Metzler matrix A, and Friedland [16, Theorem 4.1] showed that for irreducible A it is affine only along $\mathbf{1}$. Lemma 3.2 is the inverse statement, that the map on an open set determines the principal minors of A. We did not find it in these papers.

3.2. The functional determines exactly the principal minors

Theorem 3.3 (the functional and the minors)**.** *For two irreducible generators Q and Q' on $[d]$ the following are equivalent.*

(i) $J_V = J'_V$ *on* Δ_V*.*

(ii) $\lambda_V = \lambda'_V$ *on a nonempty open set of fields.*

(iii) $-Q$ *and* $-Q'$ *have the same principal minors.*

If they hold, then $\lambda_F = \lambda'_F$ *on* $\mathbb{R}^F$ *and* $J_F = J'_F$ *on* Δ_F *for every face* F*, and* F *is irreducible for* Q *exactly when it is irreducible for* Q'*.*

Proof. (i) implies (ii). By Proposition 2.3(b), $\lambda_V(v) = \min_{\Delta_V}(J_V - v \cdot \alpha) = \lambda'_V(v)$ for every v.

(ii) implies (iii). This is Lemma 3.2 with $F = V$ and $c = 0$.

(iii) implies (i). Equal principal minors give $R'_F = R_F$ for every face F, since the minors of $-Q_F$ are among those of $-Q$. By Lemma 3.1, F is irreducible for Q exactly when R_F is irreducible, which is the last claim. On an irreducible face $\lambda'_F(v)$ and $\lambda_F(v)$ are the smallest real roots of the same polynomial $t \mapsto R_F(v + t\mathbf{1})$, so they agree. For a reducible face, the factors of $R_F = R'_F$ given by Lemma 3.1 are irreducible, so by unique factorization the components of G_F have the same vertex sets for both generators, and $\lambda_F(v)$, the least $\lambda_C(v)$ over the components, agrees too. By Proposition 2.3(b), J_F is the Legendre transform of $v \mapsto -\lambda_F(v)$ on Δ_F°, so $J'_F = J_F$ there for irreducible F. On the boundary we use that J_F is convex and lower semicontinuous, as a supremum of affine functions of α. For $\alpha \in \Delta_F$ and $\beta \in \Delta_F^\circ$ the points $\alpha_t = (1-t)\alpha + t\beta$ with $t \in (0, 1]$ are interior. Convexity gives $\limsup_{t \downarrow 0} J_F(\alpha_t) \le J_F(\alpha)$, and lower semicontinuity the reverse inequality. So $J_F(\alpha) = \lim_{t \downarrow 0} J_F(\alpha_t)$, likewise for J'_F, and $J'_F = J_F$ on Δ_F. With $F = V$ this is (i). For a reducible face, Proposition 2.3(a) gives $J_F = J_V = J'_V = J'_F$ on Δ_F. □

Remark 3.4 (the survival probability). A finer observation than J_V gives the same minors more directly. Let v be a field and let the start have full support. The weighted mass $\sum_n P_N(n) e^{Tv \cdot n/N}$ is the probability that the chain survives when it is killed at each visit to i with probability $1 - e^{Tv_i/N}$, for $v \le 0$. As $N \to \infty$ at fixed T it tends to $\mu^\top e^{T(Q + \operatorname{diag} v)}\mathbf{1}$. For any generator Q' the limit $-T^{-1} \log \mu^\top e^{T(Q' + \operatorname{diag} v)}\mathbf{1}$ as $T \to \infty$ is $\lambda'_V(v) = -\varrho(Q' + \operatorname{diag} v)$, the least exit rate of the components of Q', because the start and the vector $\mathbf{1}$ are positive. So the survival probabilities on all fields give λ'_V on $\mathbb{R}^d$. Suppose that they agree with those of the irreducible Q. Then Q' is irreducible. Otherwise $\lambda'_V = \lambda'_C$ for some component $C \ne V$ on an open set of fields, and there λ'_V does not depend on v_j for $j \notin C$, while $\partial \lambda_V / \partial v_j = -\alpha_V^*(v)_j < 0$ by Proposition 2.3(b). Now Lemma 3.2 applies to Q and the irreducible Q' and gives every principal minor at once, with no descent through the faces.

3.3. The generators with given minors

Without reversibility the exit rates and the edge products $q_{ij}q_{ji}$ do not determine Q, since Q^* has the same exit rates, the same products and the same stationary law. The next theorem describes all generators with the principal minors of $-Q$.

Let C be a Z-matrix, that is $-C$ is Metzler, and let X be a cut of C with $Y = V \setminus X$. If C is irreducible, the blocks $C[X, Y]$ and $C[Y, X]$ are nonzero with nonpositive entries and rank one, so $C[X, Y] = -ab^\top$ and $C[Y, X] = -cu^\top$ with $a, b, c, u \ge 0$. The cut-transpose of C across X is

$$C = \begin{pmatrix} C_{XX} & -ab^\top \\ -cu^\top & C_{YY} \end{pmatrix} \mapsto C' = \begin{pmatrix} C_{XX} & -ac^\top \\ -bu^\top & C_{YY}^\top \end{pmatrix}. \tag{4}$$

Another choice of the factors changes C' by a positive diagonal similarity $\operatorname{diag}(\alpha I_X, I_Y)$. Transposition commutes with this operation. The transpose $C^\top$ has the same cuts, and with the factors (u, c, b, a) in place of (a, b, c, u) its cut-transpose across X is $(C')^\top$, as (4) shows by inspection.

Definition 3.5 (generator points and the cut-transpose orbit)**.** Let Q be an irreducible generator and C an irreducible Z-matrix with the principal minors of $-Q$. A positive diagonal matrix D such that $-D^{-1}CD$ is a generator exists and is unique up to a factor (step (N) of the proof of Corollary 3.6), and $-D^{-1}CD$ is the *generator point* of C. The *cut-transpose orbit* of $-Q$ is the set of matrices obtained from $-Q$ or from $-Q^{\top}$ by finitely many cut-transposes, each across a cut of the current matrix, taken up to positive diagonal similarity.

If Q has no cut, the orbit consists of the classes of $-Q$ and $-Q^{\top}$, whose generator points are Q and Q^{*}.

Corollary 3.6 (the generator fiber)**.** *Let Q be an irreducible generator.*

(a) Every generator Q' with the principal minors of $-Q$ is irreducible and has the spectrum of Q. The set of such generators is finite, and it contains Q and its time reversal Q^{*}.

(b) If Q has no cut, that is, for every partition $V = X \cup Y$ with $|X|, |Y| \geq 2$ one of the blocks $Q[X, Y]$, $Q[Y, X]$ has rank at least 2, then the set in (a) *is $\{Q, Q^{*}\}$. It is a single point exactly when Q is reversible.*

(c) In general the set in (a) *is the set of generator points of the cut-transpose orbit of $-Q$ (Definition 3.5). Every member of the orbit is an irreducible Z-matrix with the principal minors of $-Q$, and fewer than $2d$ cut-transposes reach every point of the set. If Q is reversible, the set is $\{Q\}$.*

(d) If $d \leq 3$, the set in (a) *is $\{Q, Q^{*}\}$ with no condition.*

Clause (c) needs the principal minors of $-Q$. For a general irreducible Z-matrix no diagonal similarity need produce a generator. The matrix $3I - P$ with P irreducible and stochastic is nonsingular, while the negative of every generator has the eigenvalue 0, and similarity preserves the spectrum.

What is borrowed. This is a corollary of the theory of the principal-minor map. The matrix theorems used are Loewy [23, Theorem 1] for order $n \geq 4$ without a cut, Hartfiel and Loewy [18, Theorem 3] for $n \leq 3$, combined in [7, Lemma 2.8], Al Ahmadieh [1, Theorem 4.1 and Lemma 4.5] for the converse, and Chatterjee, Ghosh, Gurjar and Raj [7, Theorem 1.1] for the whole fiber. So (b), (d) and the reversible case of (c) follow from them together with the normalization (N) below. What we add is (N), a short translation to generators, and its consequences, the finiteness, the irreducibility of every member, and the blockwise description of Proposition 3.9.

Proof. We first record 2 facts used in (a) to (d).

(N) The generator point. Let C be an irreducible Z-matrix with the principal minors of $-Q$. The characteristic polynomial $\det(tI - C) = \sum_{s=0}^{d} (-1)^{s} t^{d-s} E_s(C)$, with E_s the sum of the principal minors of order s, is that of $-Q$. So $-C$ has the spectrum of Q, whose eigenvalue of largest real part is the simple eigenvalue 0. The theorem of Perron and Frobenius for the irreducible Metzler matrix $-C$ gives a vector $x > 0$, unique up to a factor, with $Cx = 0$. With $D = \operatorname{diag} x$ the matrix $-D^{-1}CD$ has off-diagonal entries $-c_{ij}x_j/x_i \geq 0$ and zero row sums, so it is an irreducible generator. If D' is real, diagonal and invertible and $-D'^{-1}CD'$ has zero row sums, then $C(D'\mathbf{1}) = 0$, so D' is a multiple of D and gives the same matrix.

So the generator point of Definition 3.5 exists and is unique. For $C = -Q$ it is Q, and for $C = -Q^\top$, whose null vector is π, it is Q^*.

(R) Real similarities. If $-Q' = E(-Q)E^{-1}$ or $-Q' = E(-Q)^\top E^{-1}$ for a complex invertible diagonal E, then E may be taken real. In the first case $q'_{ij} = e_i q_{ij}/e_j$, so $e_i/e_j = q'_{ij}/q_{ij}$ is real on every edge of the graph of Q. That graph is strongly connected, so every ratio e_i/e_1 is a product of such ratios and is real, and E/e_1 is a real diagonal similarity. The second case is the same with q_{ji} in place of q_{ij}.

(a) Let Q' be a generator with the principal minors of $-Q$, and suppose that $-Q'$ is reducible. Then there is a proper nonempty $A \subset V$ with $Q'[V \setminus A, A] = 0$, and $\det(-Q') = \det(-Q'_A)\det(-Q'_{V\setminus A})$. Both factors are proper principal minors, so they equal $\det(-Q_A)$ and $\det(-Q_{V\setminus A})$. By Lemma 2.2 every irreducible component of a proper face has a positive exit rate, so every eigenvalue of $-Q_A$ has positive real part and $\det(-Q_A) > 0$, and likewise $\det(-Q_{V\setminus A}) > 0$. This contradicts $\det(-Q') = \det(-Q) = 0$. So Q' is irreducible, and the same argument applies to every real matrix with the principal minors of $-Q$. This also follows from Lemma 3.1, since such a matrix has the polynomial R_V of $-Q$. The spectra agree by (N). The time reversal Q^* is a generator, since $Q^*\mathbf{1} = D_\pi^{-1}Q^\top\pi = 0$ and its off-diagonal entries are $\pi_j q_{ji}/\pi_i \geq 0$, and $-Q^* = D_\pi^{-1}(-Q)^\top D_\pi$ has the principal minors of $-Q$, since transposition and diagonal similarity preserve them. Finiteness follows from (c).

(b) For $d \leq 3$ the claim is (d). Let $d \geq 4$. Loewy's theorem [23, Theorem 1] applies to the irreducible matrix $-Q$ without a cut, with no condition on $-Q'$. It gives an invertible diagonal E with $-Q' = E(-Q)E^{-1}$ or $-Q'^\top = E(-Q)E^{-1}$, and by (R) we may take E real. In the first case (N) with $C = -Q$ gives $Q' = Q$, and in the second case $Q' = E^{-1}Q^\top E$ and (N) with $C = -Q^\top$ gives $Q' = Q^*$. Finally $Q^* = Q$ exactly when $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j, that is when Q is reversible.

(c) Let C and X be as in (c). The cut-transpose C' is a Z-matrix. It has the same principal minors as C. Indeed, for $U \subseteq V$ write $U_X = U \cap X$ and $U_Y = U \cap Y$. If U_X or U_Y is empty, both minors agree since a determinant is invariant under transposition. Otherwise replace C_{XX} by $C_{XX} + sI$ with s large, so that the block of U_X is invertible. The Schur complement gives

$$\det C_U = \det C_{U_X} \det\big(C_{U_Y} - \kappa\, c_{U_Y} b_{U_Y}^\top\big), \qquad \det C'_U = \det C_{U_X} \det\big(C_{U_Y}^\top - \kappa\, b_{U_Y} c_{U_Y}^\top\big),$$

with the same scalar $\kappa = u_{U_X}^\top C_{U_X}^{-1} a_{U_X}$. The two right sides are equal by transposition. Both sides are polynomials in s that agree for large s, so they agree at $s = 0$. By the argument of (a), C' is irreducible, and by (N) it has exactly one generator point.

Now let Q' be a generator in the fiber, irreducible by (a). The theorem of Chatterjee, Ghosh, Gurjar and Raj [7, Theorem 1.1] gives a sequence $-Q = C_0, C_1, \ldots, C_k$ with $k < 2d$, each C_{s+1} a cut-transpose of C_s across a cut of C_s [7, Definition 2.10], such that $-Q'$ is diagonally similar to C_k or to $C_k^\top$. Their cut-transpose uses a particular rank-one factorization, and another choice changes it by a real diagonal similarity $\mathrm{diag}(\alpha I_X, I_Y)$ [7, Remark 2.11]. A real invertible diagonal similarity of C by $\mathrm{diag}(E_X, E_Y)$ changes the cut-transpose by $\mathrm{diag}(E_X, E_Y^{-1})$, as a short computation with the blocks shows, and does not change the cuts. So, up to real invertible diagonal similarity, which (N) accepts, every C_s is obtained by our cut-transposes, and Q' is the generator point of C_k or of $C_k^\top$. Since transposition commutes with cut-transposes, $C_k^\top$ is reached from $-Q^\top$ by the same sequence, so $-Q'$ lies in the orbit

of Definition 3.5. Conversely every generator point of the orbit lies in the fiber, by the first paragraph of (c) and the invariance of the minors under transposition. There are fewer than 2^d cuts at each step and fewer than $2d$ steps, so the fiber is finite, which completes (a).

If Q is reversible, $S = D_{\sqrt{\pi}}(-Q)D_{\sqrt{\pi}}^{-1}$ is an irreducible symmetric Z-matrix, and a symmetric matrix is its own cut-transpose. Indeed, if $S[X,Y] = -ab^{\top}$, then $S[Y,X] = -ba^{\top}$, so we may take $c = b$ and $u = a$. By the rule for diagonal similarities above, the cut-transpose of a matrix diagonally similar to S is again diagonally similar to S, and so is the transpose $E^{-1}SE$ of ESE^{-1}. Since $C_0 = -Q$ is diagonally similar to S, so is every C_s. Hence $-Q'$ is diagonally similar to $-Q$, and (N) with $C = -Q$ gives $Q' = Q$. See also Engel and Schneider [13, Theorem 3.5] and Al Ahmadieh [1, Lemma 6.1 and Theorem 6.2].

(d) For $d = 2$ a generator is fixed by q_1 and q_2, which are principal minors. Let $d = 3$ and let Q' be in the fiber, so Q' is irreducible by (a). The theorem of Hartfiel and Loewy [18, Theorem 3], as restated in [7, Lemma 2.8], says that for an irreducible matrix A of order 2 or 3 over a field, every matrix with the principal minors of A is diagonally similar to A or to $A^{\top}$ over that field. So $-Q' = E(-Q)E^{-1}$ or $-Q' = E(-Q)^{\top}E^{-1}$ with E diagonal and invertible, real by (R), and (N) gives $Q' = Q$ or $Q' = Q^*$, as in (b). □

A cut-transpose need not be Q or Q^*. Take $d = 4$, $X = \{1,2\}$ and

$$Q = \begin{pmatrix} -4.7 & 0.7 & 1 & 3 \\ 1.3 & -9.3 & 2 & 6 \\ 2 & 2 & -4.4 & 0.4 \\ 1 & 1 & 2 & -4 \end{pmatrix},$$

whose blocks $Q[X,Y] = (1,2)^{\top}(1,3)$ and $Q[Y,X] = (2,1)^{\top}(1,1)$ have rank one. To 4 decimals, the generator point of the cut-transpose of $-Q$ across X is

$$Q' = \begin{pmatrix} -4.7 & 0.7 & 2.3810 & 1.6190 \\ 1.3 & -9.3 & 4.7619 & 3.2381 \\ 0.84 & 0.84 & -4.4 & 2.72 \\ 1.8529 & 1.8529 & 0.2941 & -4 \end{pmatrix}.$$

Its entries are rational, for instance $q'_{13} = 50/21$, $q'_{41} = 63/34$ and $q'_{43} = 5/17$. It has the 15 principal minors of $-Q$, with $Q'_X = Q_X$ and $Q'_Y = Q^*_Y$, but $Q' \neq Q^*$, since $q^*_{12} = 351/530 \approx 0.6623$.

Remark 3.7 (three states by hand). For $d = 3$ the minors give the exit rates, the products $q_{ij}q_{ji}$, and the sum and product of the two oriented 3-cycle products $\omega = q_{12}q_{23}q_{31}$ and $\omega' = q_{13}q_{32}q_{21}$. So Q' has the cycle products of Q or of Q^*, hence differs from it by a potential, and (N) gives (d). Example 5.4 of Kim and Landim [20] is the case $\omega = 1$, $\omega' = 0$.

3.4. What the exact law adds

The law of the counts is finer than J_V, and for a generic start it separates the fiber. From the stationary start it cannot separate Q from Q^* at all.

Remark 3.8 (the stationary start). Let Q be irreducible with stationary law π. With the start π, the law of the composition is the same under Q and under Q^*, for every N and every T with

$Tq_{\max} \le N$, and so is the weighted law $P_N(n)e^{Tv\cdot n/N}$ for every field v. This is path reversal, and we include the proof only to fix the discrete-time form. The matrix $P = I + (T/N)Q$ has stationary law π, and $P^* = D_\pi^{-1}P^\top D_\pi = I + (T/N)Q^*$ is stochastic, since Q^* has the exit rates of Q. For a path $x_1 \cdots x_N$ we have $\pi_{x_1}\prod_t P(x_t, x_{t+1}) = \pi_{x_N}\prod_t P^*(x_{t+1}, x_t)$. Reversal is a bijection of the paths that keeps the composition. Summing over the paths with composition n gives the claim, since the factor $e^{Tv\cdot n/N}$ depends only on n.

For a face F that is irreducible for Q and a field v, the face maximum has the constant $K_F(v)$ of Definition 2.6. Let r and l be the right and left Perron vectors of $Q_F + \operatorname{diag} v_F$ with $l \cdot r = 1$. By Proposition 2.7,

$$K_F(v) = \frac{a_F(v)}{(2\pi)^{(k-1)/2}\sqrt{\det \Sigma_{\alpha_F^*(v)}}}, \qquad a_F(v) = (\mu_F \cdot r)(l \cdot \mathbf{1}), \tag{5}$$

and we call $a_F(v)$ the amplitude.

Proposition 3.9 (the amplitudes)**.** *Let Q be irreducible, let $Q' \ne Q$ be a generator in the fiber of Corollary 3.6, and let F be a face that is irreducible for Q.*

(a) F is irreducible for Q', the two generators have the same λ_F, α_F^*, J_F and $\det \Sigma_{\alpha_F^*(v)}$, and $K'_F(v)/K_F(v) = a'_F(v)/a_F(v)$.

(b) There is a finite union $\mathcal{Z}$ of proper linear subspaces of $\mathbb{R}^V$ with the following property. For every start μ of full support outside $\mathcal{Z}$ and every two distinct generators Q' and Q'' in the fiber, $K'_V(v) \ne K''_V(v)$ for all v outside a closed nowhere dense set of fields. That set always contains $v = 0$, where every generator has $r = \mathbf{1}$, $l = \pi$ and $a_V(0) = 1$. If Q is not reversible, then $\pi \in \mathcal{Z}$.

(c) If Q' is the generator point of one cut-transpose of $-Q$ across X, then for every start $K'_F = K_F$ on every face $F \subseteq X$ and $K'_F = K^*_F$ on every face $F \subseteq Y$, where K^* belongs to Q^*.

Proof. (a) The first claims are Theorem 3.3. So $\alpha_F^*(v)$ and, by Proposition 2.7, $\det \Sigma$ at that point agree for Q and Q', and (5) gives the ratio.

(b) Take $F = V$ and put $g(v) = r(v)\,(l(v)\cdot\mathbf{1})$, which does not depend on the scale of r, so that $a_V(v) = \mu \cdot g(v)$. The map g is real-analytic on $\mathbb{R}^V$. Let W be the span of the vectors $g'(v) - g(v)$ over $v \in \mathbb{R}^V$. If $\mu \notin W^\perp$, the analytic function $a'_V - a_V = \mu \cdot (g' - g)$ is not identically zero, so its zero set is closed and nowhere dense. We show that $W \ne 0$. Suppose $g' = g$. Then $r'(v)$ is a multiple of $r(v)$, and both matrices have the eigenvalue $\lambda_V(v)$, so $(Q' - Q)r(v) = 0$ for every v. Take $v = te_j$ with $t \to \infty$. Then $\lambda_V \le q_j - t \to -\infty$, and row $i \ne j$ of the eigenvalue equation gives $(q_i - \lambda_V)r_i \le q_{\max}\max r$, so $r(te_j)/\max r \to e_j$. Hence $(Q' - Q)e_j = 0$ for every j and $Q' = Q$. So for $Q' \ne Q$ the space $W^\perp$ is proper. Every member Q'' of the fiber is irreducible and has the same fiber as Q, by Corollary 3.6(a), so the same argument with Q'' in place of Q gives a proper space $W^\perp_{Q'',Q'}$ for every ordered pair of distinct members. Let $\mathcal{Z}$ be the union of these finitely many spaces. At $v = 0$ every generator has $r = \mathbf{1}$ and $l = \pi$, so $a_V(0) = \mu(V) = 1$. For the time reversal, $-Q^* - \operatorname{diag} v = D_\pi^{-1}(-Q - \operatorname{diag} v)^\top D_\pi$, so $r^* = D_\pi^{-1}l$ and $l^* = D_\pi r$, with $l^* \cdot r^* = 1$. At $\mu = \pi$ this gives $a_V^*(v) = (\mathbf{1}\cdot l)(\pi \cdot r) = a_V(v)$ for all v, so $\pi \in \mathcal{Z}$ when $Q^* \ne Q$, in line with Remark 3.8.

(c) Let $C = -Q$ and let C' be its cut-transpose across X, with the factors of (4), and put $s_0 = (b \cdot \mathbf{1})/(c \cdot \pi_Y)$. The rows X of $Q\mathbf{1} = 0$ give $-Q_X\mathbf{1} = (b \cdot \mathbf{1})a$, the columns Y of $\pi^\top Q = 0$ give $-Q_Y^\top \pi_Y = (\pi_X \cdot a)b$, and summing the columns X of $\pi^\top Q = 0$ gives $(\pi_X \cdot a)(b \cdot \mathbf{1}) = (\pi_Y \cdot c)(u \cdot \mathbf{1})$. With these identities, $x = (\mathbf{1}_X, s_0\pi_Y)$ satisfies $C'x = 0$. So $Q' = -D_x^{-1}C'D_x$, and its killed blocks are $Q'_X = Q_X$ and $Q'_Y = D_{\pi_Y}^{-1}Q_Y^\top D_{\pi_Y} = Q_Y^*$. The constant $K_F(v)$ of a face depends only on Q_F and μ_F, which gives the claim. □

Proposition 3.9 concerns the law, not the labels of Section 4 (see the open questions in Section 6).

4. Modal supports

This section proves Theorem B. The observable is artificial, and we say so first. The survival probability under calibrated killing is a finer observation, it is easier to sample, and it gives every principal minor at once (Remark 3.4). We know of no experiment in which the modal support is observed and the survival probability is not. We study the label because the forward theory of [25] describes the law of the counts through it, and so does any exact optimizer that reports only a maximizer. Theorem B says that its first-order form loses nothing that J_V keeps.

4.1. The killed chain and two labels

Fix a field $v \in \mathbb{R}^d$ and $\tau > 0$, put $T = \tau L$, and weight each composition by $e^{Tv\cdot n/N}$. If $v \le 0$, the weight of a path is the probability that the chain survives when it is killed at each visit to i with probability $1 - e^{Tv_i/N}$. So the maximizers of $P_N(n)e^{Tv\cdot n/N}$ are the modes of the composition of the surviving chain. On a face the killing shifts the diagonal, $Q_F + \operatorname{diag} v_F$ in place of Q_F. The shift $v \mapsto v + c\mathbf{1}$ multiplies the weighted law by e^{Tc} and moves every score $\Phi_{\tau,v}(F)$ by $-\tau c$, so it changes no winner and every field may be taken negative. The field and the time scale are calibrated, that is, known in the units of Q.

Proposition 4.1 (the controlled score law)**.** *Fix Q, a start μ with full support, a field v and $\tau > 0$, and let $T = \tau L$. For every admissible face F, $\log M_F(T, v) \le -\Phi_{\tau,v}(F)\, L + O(\log L)$, with equality up to $O(\log L)$ if F is irreducible. A reducible admissible face F has $\Phi_{\tau,v}(F) \ge \Phi_{\tau,v}(C)+1$ for one of its strongly connected components C. Hence if F is the unique minimizer of $\Phi_{\tau,v}$ over the admissible faces, then F is irreducible and the support of every maximizer of the weighted law is F for all large N.*

Proof. For large N we have $T_0 \le \tau L \le N^{1/3}$. Theorem 2.5(a) gives the upper bound, and Theorem 2.8 the equality, since an irreducible face satisfies (H) by Lemma 2.1(a). Let F be reducible and admissible. Then $\lambda_F(v)$ is the least $\lambda_C(v)$ over the components, and a component C where it is attained is irreducible with $|C| \le |F| - 1$, so $\Phi_{\tau,v}(F) = \Phi_{\tau,v}(C) + |F| - |C|$. If F is the unique minimizer and $\varepsilon > 0$ is its least gap to the other admissible faces, their weighted maxima are at most $N^{-\Phi_{\tau,v}(F)-\varepsilon+o(1)}$, that of F is at least $N^{-\Phi_{\tau,v}(F)-o(1)}$, and a face that is not admissible carries no mass. □

Let $\Omega = (0, \infty) \times \mathbb{R}^d$ and fix a start of full support. We distinguish two labels. The *finite-horizon label* $\mathcal{W}_N(\omega)$ at $\omega = (\tau, v)$ is the set of supports of the maximizers of $n \mapsto P_N(n)e^{Tv\cdot n/N}$. The *first-order modal-support map* is defined on the strict set $\mathcal{S}_Q$ of the points ω at which $\Phi_{\tau,v}$ has a unique minimizer over the admissible faces, and $W_Q(\omega)$ is that minimizer.

Lemma 4.2 (strict regions)**.** *Let Q be a generator and μ a start of full support.*

(a) $\mathcal{S}_Q$ is open and dense in Ω, and W_Q is locally constant on it.

(b) At every $\omega \in \mathcal{S}_Q$ the face $W_Q(\omega)$ is irreducible, and $\mathcal{W}_N(\omega) = \{W_Q(\omega)\}$ for all $N \geq N_0(\omega)$.

(c) $\mathcal{S}_Q$ and W_Q are invariant under $(\tau, v) \mapsto (\tau, v + c\mathbf{1})$.

Proof. (b) is Proposition 4.1 and (c) is the shift rule. For (a), the scores are continuous in (τ, v) and there are finitely many faces, so $\mathcal{S}_Q$ is open and W_Q locally constant. A reducible face never attains the least score, so the complement of $\mathcal{S}_Q$ lies in the union of the closed sets Z_{FH} where $\Phi_{\tau,v}(F) = \Phi_{\tau,v}(H)$, over pairs of distinct irreducible faces. If $|F| \neq |H|$, the difference vanishes for at most one τ at fixed v. If $|F| = |H|$, pick $j \in F \setminus H$. Then $\lambda_H(v)$ does not depend on v_j, while $\partial\lambda_F/\partial v_j = -\alpha_F^*(v)_j < 0$ by Proposition 2.3(b). So each Z_{FH} has empty interior, and so has their finite union. □

By Lemma 4.2(b) the finite-horizon labels at all N determine W_Q on its strict set. They may determine more, since the positions of the label changes at finite N involve the constants $K_F(v)$ (Section 6). Theorem B concerns W_Q only, and only for W_Q do we say *exactly*.

Definition 4.3 (controlled boundaries)**.** Fix a start μ of full support. Two generators Q and Q' have the same controlled boundaries if $W_Q = W_{Q'}$ on $\mathcal{S}_Q \cap \mathcal{S}_{Q'}$.

The set $\mathcal{S}_Q \cap \mathcal{S}_{Q'}$ is open and dense, and both maps are locally constant on it. So Q and Q' fail to have the same controlled boundaries exactly when there is a nonempty open set of ω on which both have a strict winner and the winners differ. By Lemma 4.2(c) it suffices to compare the maps on $(0, \infty) \times (-\infty, 0)^d$, where every field is a killing rate. Ties are never observed and never used.

4.2. Suppression and exposure

For a Metzler matrix B and a positive vector u, the Perron root of B is at most $\max_i (Bu)_i/u_i$, by the bound of Collatz and Wielandt [2].

Lemma 4.4 (suppression)**.** *Let Q be a generator, F a nonempty face, $\mathcal{K} \subset \mathbb{R}^F$ compact and $\varepsilon > 0$. There is $\bar{v}_0$, depending only on Q, F, $\mathcal{K}$ and ε, such that for every field v with $v_F \in \mathcal{K}$ and $v_j \leq -\bar{v}_0$ for all $j \notin F$, and every face $G \not\subseteq F$, we have $\lambda_G(v) \geq \lambda_{G\cap F}(v) - \varepsilon$ if $G \cap F \neq \emptyset$, and $\lambda_G(v) \geq \bar{v}_0$ if $G \cap F = \emptyset$. Every larger value of $\bar{v}_0$ also serves.*

Proof. Put $P = G \cap F$, $S = G \setminus F$ and $B = Q_G + \operatorname{diag} v_G$. If $P = \emptyset$, every row sum of B is at most $-\bar{v}_0$, and the bound with $u = \mathbf{1}$ gives $\lambda_G(v) \geq \bar{v}_0$. Let $P \neq \emptyset$, $\varrho_P(v) = -\lambda_P(v)$, and let $\mathbf{J}_P$ have ones off the diagonal of P. For $\zeta > 0$ the Perron root of the irreducible matrix $Q_P + \operatorname{diag} v_P + \zeta \mathbf{J}_P$ decreases to $\varrho_P(v)$ as $\zeta \downarrow 0$, continuously in v_P. By Dini's theorem one ζ makes it at most $\varrho_P(v) + \varepsilon/2$ for all $v_F \in \mathcal{K}$. Let x be its Perron vector with $\max x = 1$, let $x_* > 0$ be the least entry of x over $\mathcal{K}$, and test B with $u = x$ on P and $u = y_0 = \varepsilon x_*/(2q_{\max})$ on S. For $p \in P$, since $\mathbf{J}_P x \geq 0$, $(Bu)_p \leq (\varrho_P + \frac{\varepsilon}{2})x_p + y_0 q_{\max} \leq (\varrho_P + \varepsilon)x_p$. For $s \in S$, the off-diagonal entries of row s inside S sum to at most q_s, so $(Bu)_s \leq q_{\max} + y_0 v_s \leq q_{\max} - y_0\bar{v}_0 \leq (\varrho_P + \varepsilon)y_0$ once $\bar{v}_0 \geq q_{\max}/y_0 + \max_{\mathcal{K}} |\varrho_P| + \varepsilon$. There are finitely many P and S, so one $\bar{v}_0$ serves for all, and so does every larger value. □

For an irreducible face F with $k = |F| \geq 2$ and an irreducible face $H \subsetneq F$, singletons included, put

$$\rho_H(v) = \frac{k - |H|}{\lambda_H(v) - \lambda_F(v)}, \qquad v \in \mathbb{R}^F.$$

The denominator is positive by Lemma 2.2, ρ_H is real-analytic, and $\Phi_{\tau,v}(F) < \Phi_{\tau,v}(H)$ exactly when $\tau > \rho_H(v)$. So as τ decreases, the first proper subface to tie with F maximizes $\rho_H(v)$.

Lemma 4.5 (exposure by descent)**.** *Let Q be irreducible with a start of full support, and let F be an irreducible face with $|F| \geq 2$. There is a closed nowhere dense set $\mathcal{E} \subset \mathbb{R}^F$ such that every $v^0 \in \mathbb{R}^F \setminus \mathcal{E}$ has a unique partner G, the irreducible face $G \subsetneq F$ that maximizes $\rho_H(v^0)$. Put $\tau_1 = \rho_G$. There are an open neighbourhood $\mathcal{U}$ of v^0 with compact closure and numbers $\delta > 0$ and $\bar{v}_0$, depending on Q, F and v^0, such that on the open set*

$$O = \big\{(\tau, v) \in \Omega : \ v_F \in \mathcal{U}, \ v_j < -\bar{v}_0 \text{ for } j \notin F, \ |\tau - \tau_1(v_F)| < \delta\big\}$$

the following hold.

(i) The points of O with $\tau > \tau_1(v_F)$ lie in $\mathcal{S}_Q$ with $W_Q = F$, and those with $\tau < \tau_1(v_F)$ lie in $\mathcal{S}_Q$ with $W_Q = G$.

(ii) On the hypersurface $\tau = \tau_1(v_F)$ we have $\lambda_F(v) = \lambda_G(v) - (|F| - |G|)/\tau_1(v_F)$.

For $F = V$ the condition on v_j is empty.

Proof. Generic partners. Let $H_1 \neq H_2$ be irreducible proper subfaces of F with some $j \in H_1 \setminus H_2$, and let $v = te_j$ with $t \to \infty$. Then λ_{H_2} stays fixed, $-t \leq \lambda_F \leq \lambda_{H_1} \leq q_j - t$, so $\rho_{H_2} \to 0$ while $\rho_{H_1} \geq (k - |H_1|)/q_j > 0$. So $\rho_{H_1} - \rho_{H_2}$ is a nonzero real-analytic function, and the set $\mathcal{E}$ where the maximum of ρ_H is attained twice is closed and nowhere dense.

The boundary. Let G be the partner of $v^0 \notin \mathcal{E}$. Choose $\delta > 0$ with $3\delta < \tau_1(v^0)$ and $\rho_H(v^0) < \tau_1(v^0) - 3\delta$ for every other irreducible $H \subsetneq F$, and take for $\mathcal{U}$ a ball around v^0 on whose closure these hold with 2δ. Let $\tau_\pm$ be the largest and least τ on the closure of O, so $\tau_- > \delta$. There $\Phi(H) - \Phi(F) = (\lambda_H - \lambda_F)(\tau - \rho_H) \geq \kappa > 0$ for every such H, while $\Phi(F) - \Phi(G) = (\lambda_G - \lambda_F)(\tau_1(v_F) - \tau)$ changes sign on the hypersurface, and a reducible face loses to a component. So among the faces inside F the least score is attained only by F above and only by G below. For the faces $G' \not\subseteq F$ apply Lemma 4.4 with the closure of $\mathcal{U}$ and $\varepsilon = 1/(2\tau_+)$. If $P = G' \cap F \neq \emptyset$ and C is a component of P with $\lambda_C(v) = \lambda_P(v)$, then $\Phi(G') \geq |P| + \tau\lambda_P(v) - \frac{1}{2} \geq \Phi(C) + \frac{1}{2}$. Enlarge $\bar{v}_0$ until $\tau_-\bar{v}_0$ exceeds the largest score of F on the closure of O by $\frac{1}{2}$. Then a face with P empty also scores $\frac{1}{2}$ above the least score, which proves (i). The tie $\Phi(F) = \Phi(G)$ on the hypersurface is (ii). □

Theorem 4.6 (the modal supports and the minors)**.** *Let Q and Q' be irreducible generators and let the start have full support. They have the same controlled boundaries if and only if $-Q$ and $-Q'$ have the same principal minors. If Q is irreducible and Q' is any generator with the controlled boundaries of Q, then Q' is irreducible.*

Proof. Equal minors. By Theorem 3.3, Q and Q' have the same irreducible faces and the same λ_F on each. With a start of full support every irreducible face is admissible, and a reducible admissible face scores above one of its components. So $\mathcal{S}_Q = \mathcal{S}_{Q'}$ and $W_Q = W_{Q'}$.

Irreducibility. At a fixed field, V is the strict winner for Q once τ is large (Lemma 2.2), on an open set of Ω. That set meets the dense set $\mathcal{S}_{Q'}$, where $W_{Q'} = V$. A reducible face is never a strict winner, so Q' is irreducible.

Singletons. Fix $i \neq 1$ and let $\bar{\tau} = 1/(2\max(q_{\max}, q'_{\max}))$. Let I be a compact interval in $(0, \bar{\tau}]$ and J a compact interval with $q_i - q_1$ and $q'_i - q'_1$ in its interior. Consider the open set of (τ, v) with τ in the interior of I, $|v_1| < 1$, $v_i - v_1$ in the interior of J and $v_j < -\bar{v}_0$ for $j \notin \{1, i\}$. Here $\bar{v}_0$ is the larger threshold of Lemma 4.4 for Q and Q', with $F = \{1, i\}$, the compact set $[-1, 1] \times ([-1, 1] + J)$ of (v_1, v_i) and $\varepsilon = 1/(4\bar{\tau})$, enlarged as in Lemma 4.5. The pair $\{1, i\}$, if it is irreducible, has exit rate at least $\min(q_1 - v_1, q_i - v_i) - \sqrt{q_{1i}q_{i1}}$ and $\sqrt{q_{1i}q_{i1}} \leq q_{\max}$, so it scores at least $\frac{1}{2}$ above the better singleton, for both generators. Hence the strict winners are $\{1\}$ and $\{i\}$ on the two sides of $v_i - v_1 = q_i - q_1$ for Q, and of $v_i - v_1 = q'_i - q'_1$ for Q'. If the hyperplanes differed, the slab between them would carry different winners. So $q'_i - q'_1 = q_i - q_1$, and with $c = q'_1 - q_1$ we get $\lambda'_{\{i\}} = \lambda_{\{i\}} + c$ for every i.

The descent. We show by induction on $|F|$ that every irreducible face F of Q is irreducible for Q' with $\lambda'_F = \lambda_F + c$ on $\mathbb{R}^F$. Let $|F| \geq 2$, choose $v^0 \notin \mathcal{E}$ with partner G, and take O from Lemma 4.5. On the dense set $O \cap \mathcal{S}_{Q'}$ we have $W_{Q'} = W_Q$, so F and G are irreducible for Q', and by continuity $\Phi'(F) = \Phi'(G)$ on the hypersurface. With (ii) and the induction hypothesis,

$$\lambda'_F(v) = \lambda'_G(v) - \frac{|F| - |G|}{\tau_1(v_F)} = \lambda_G(v) + c - \frac{|F| - |G|}{\tau_1(v_F)} = \lambda_F(v) + c \qquad (v_F \in \mathcal{U}),$$

and Lemma 3.2 extends this to $\mathbb{R}^F$. For $F = V$ both generators have $\lambda_V(0) = 0$, so $c = 0$, and Lemma 3.2 gives equal principal minors. □

The descent is an algorithm. For each face choose a field at random, decrease τ from a large value, and read the first change of label and its position. Its open sets have thresholds that depend on Q and Q', so no experiment is fixed in advance, and the statement is one of identifiability.

5. Low temperature and the cycle hierarchy

We use the cycles of Freidlin and Wentzell [15] in the form of Catoni [6]. The depth of a face (Theorem 5.1) is a short consequence of Catoni's theorem on the exit time of a domain, and the prefactors (Proposition 5.3) are classical in substance [28]. Section 5.2 proves Theorem C. Its main steps are the depth of a face under an added calibrated killing (Lemma 5.9), the strict growth of depths along nested cycles (Lemma 5.10), and a comparison of the rewarded cycle with each kind of face.

5.1. Arrhenius families

Let $V_{ij} \geq 0$ for $i \neq j$ be fixed costs and let

$$q_{ij}(\eta) = e^{-V_{ij}/\eta}, \qquad \eta > 0,$$

so every off-diagonal rate is positive at every positive temperature. We write Q_η for the generator, and for a face F we write λ_F^η for the exit rate λ_F of F under Q_η, and λ_F when η

is fixed. In this section the letter h is a height. For a proper face F define its depth

$$D(F) = \lim_{\eta\downarrow 0}\bigl(-\eta \log \lambda_F^\eta\bigr),$$

whose existence is part of Theorem 5.1. Put $D(V) = \infty$. A fixed height $h > 0$ sets the time scale

$$\tau_\eta = e^{h/\eta}, \qquad T = \tau_\eta L.$$

Throughout the section we take $N \to \infty$ first at fixed η and then $\eta \downarrow 0$, and every threshold in N may depend on η. The start μ_η may depend on η but has full support and does not depend on N. Since all rates are positive, every face is irreducible, and with a full-support start every face satisfies (H) by Lemma 2.1(a). So Theorem 2.5(b) applies at every fixed η to every face. At fixed η the face maxima of Theorem 2.8 give

$$\log M_F = -\Phi_{\eta,h}(F)\, L + \frac{|F|-1}{2}\log L + O_\eta(1), \qquad \Phi_{\eta,h}(F) = |F| - 1 + e^{h/\eta}\lambda_F^\eta. \tag{6}$$

So $\Phi_{\eta,h}$ is the exponent Φ_τ of Theorem 2.5 for the generator Q_η at $\tau = e^{h/\eta}$. The constants in (6) may depend badly on η, and nothing below exchanges the two limits.

We use the cycles of Freidlin and Wentzell in Catoni's convention [6, Definition 4.2]. A nonempty proper set C is a cycle if it is a singleton or if, for all distinct $x, y \in C$, the probability of leaving C before hitting y, starting from x, is exponentially small in $1/\eta$. Its depth $\Gamma(C)$ is the exponential scale of its mean exit time, which does not depend on the starting point in C [6, Proposition 4.6], and $\Gamma(\{i\}) = \min_{j\neq i} V_{ij}$. For a reversible family write $V_{ij} = H_{ij} - U_i$ with H symmetric, so that $\pi_i(\eta) \propto e^{-U_i/\eta}$. Let $\beta(C) = \min_{i\in C, j\notin C} H_{ij}$ be the boundary height of a proper set C with at least 2 states. Then C is a cycle exactly when any 2 of its points are joined by a path whose saddles H are all less than $\beta(C)$, and then $\Gamma(C) = \beta(C) - \min_{i\in C} U_i$ [8, Definition 3.5, Propositions 3.10 and 3.15]. We call this the energy criterion. The inequality is strict, since a weak one would admit sets at a common saddle level that are not cycles.

Theorem 5.1 (the depth of a face)**.** *For every nonempty proper face F the depth $D(F)$ exists and*

$$D(F) = \max\{\Gamma(C) : C \subseteq F \text{ a cycle}\}.$$

In particular $D(C) = \Gamma(C)$ for every cycle C, and F contains a cycle C with $D(C) = D(F)$. Moreover, with $\mathcal{G}_F = (-Q_F)^{-1}$ and $M_F^{\mathrm{exit}} = \max_{i\in F} \mathbb{E}_i \tau_{F^c}$,

$$\frac{1}{M_F^{\mathrm{exit}}} \le \lambda_F^\eta \le \frac{|F|}{M_F^{\mathrm{exit}}},$$

so that $D(F) = \lim_{\eta\downarrow 0} \eta \log M_F^{\mathrm{exit}}$.

Proof. Fix η and a proper face F, and write λ_F for λ_F^η. By Lemma 2.2 with $v = 0$, $\lambda_F > \lambda_V = 0$. So $\mathcal{G}_F = \int_0^\infty e^{tQ_F}\, dt$ converges, and since $(e^{tQ_F})_{ij} = \mathbb{P}_i(X_t = j,\ t < \tau_{F^c})$ for the chain X with generator Q_η,

$$(\mathcal{G}_F)_{ij} = \mathbb{E}_i \int_0^{\tau_{F^c}} \mathbf{1}\{X_t = j\}\, dt = \mathbb{P}_i(\tau_j < \tau_{F^c})\,(\mathcal{G}_F)_{jj} \le (\mathcal{G}_F)_{jj}$$

by the strong Markov property at τ_j.

The exit-time bounds. The row sums of $\mathcal{G}_F$ are the mean exit times $\mathbb{E}_i\tau_{F^c}$. The spectral radius ρ of a nonnegative matrix is at most its largest row sum and at least each of its diagonal entries. Hence

$$\rho(\mathcal{G}_F) \le M_F^{\text{exit}} \le \operatorname{tr}\mathcal{G}_F \le |F| \max_j (\mathcal{G}_F)_{jj} \le |F|\rho(\mathcal{G}_F).$$

The Perron vector of $-Q_F$ is positive and is an eigenvector of $\mathcal{G}_F$ for $1/\lambda_F$. A positive eigenvector of a nonnegative matrix belongs to its spectral radius [2], so $\rho(\mathcal{G}_F) = 1/\lambda_F$, which gives the bounds.

The depth. Fix $\varkappa > d$ and put $P_\eta = I + Q_\eta/\varkappa$. Every rate is at most 1, so every exit rate is at most $d-1$, and P_η is stochastic with holding probabilities at least $1-(d-1)/\varkappa > 0$. Then $-\eta \log P_\eta(i,j) \to V_{ij}$ for $i \ne j$ and $-\eta \log P_\eta(i,i) \to 0$. So (P_η) has rare transitions in the sense of [6, Definition 2.1], with $\beta = 1/\eta$ and the rate function equal to V_{ij} off the diagonal and to 0 on it, and this rate function is irreducible since all rates are positive. The chain P_η visits the same states in the same order as X, so the two chains have the same cycles. The continuous chain is the discrete one run at the jump times of a Poisson clock of rate $\varkappa$, so its mean exit times are the discrete ones divided by $\varkappa$. For a domain $\mathcal{O}$ and $x \in \mathcal{O}$, Catoni's Proposition 4.2 [6] gives the limit $H_{\mathcal{O}}(x)$ of $\beta^{-1}\log$ of the mean exit time from $\mathcal{O}$ started at x. Let $\mathcal{M}(\mathcal{O})$ be the set of maximal cycles contained in $\mathcal{O}$ (his Definition 4.3), and $\Gamma(C)$ the depth of his Proposition 4.6. His Proposition 4.11 states that $\max_{x\in\mathcal{O}} H_{\mathcal{O}}(x) = \max\{\Gamma(C) : C \in \mathcal{M}(\mathcal{O})\}$ for every domain $\mathcal{O}$ other than the whole space. With $\mathcal{O} = F$ this reads

$$\lim_{\eta\downarrow 0} \eta \log M_F^{\text{exit}} = \max_{C\in\mathcal{M}(F)} \Gamma(C).$$

These are cycles of the whole chain, not of a chain restricted to F. By the exit-time bounds $-\eta\log\lambda_F$ differs from $\eta \log M_F^{\text{exit}}$ by at most $\eta\log|F|$, so $D(F)$ exists and equals this limit. Every cycle $C \subseteq F$ lies in some $C' \in \mathcal{M}(F)$, and $\Gamma(C) \le \Gamma(C')$ since $\tau_{C^c} \le \tau_{C'^c}$ pathwise. So the maximum over $\mathcal{M}(F)$ is the maximum over all cycles in F, attained at some $C \in \mathcal{M}(F)$. If F is a cycle, then $\mathcal{M}(F) = \{F\}$ and $D(F) = \Gamma(F)$. □

Theorem 5.2 (noncritical heights)**.** *Let $\mathcal{D}$ be the finite set of depths of proper cycles and let $h \notin \mathcal{D}$. Let $k(h)$ be the least size of a cycle of depth greater than h, where V counts as a cycle of infinite depth and size d. Then for all sufficiently small η, every minimizer of $\Phi_{\eta,h}$ is a cycle of size $k(h)$ and depth greater than h, and every eventual support of the global mode, in the limit $N \to \infty$ at that η, is such a cycle.*

Proof. By Theorem 5.1 the depth of every proper face lies in $\mathcal{D}$ and differs from h. If $D(F) < h$, then $e^{h/\eta}\lambda_F^\eta = e^{(h-D(F)+o(1))/\eta} \to \infty$. If $D(F) > h$, then $e^{h/\eta}\lambda_F^\eta \to 0$. The face V has the score $d-1$. Call a face deep if it is V or a proper face of depth greater than h.

A deep face of least size is a cycle. Indeed, a deep proper face F contains a cycle C with $D(C) = D(F) > h$ by Theorem 5.1, and C is deep, so $C = F$ if F has least size. Every cycle of depth greater than h is deep. So the deep faces of least size are the cycles of size $k(h)$ and depth greater than h.

There are finitely many faces, so for all sufficiently small η every face that is not deep has $\Phi_{\eta,h} > d$, and every deep face of size $k(h)$ has $\Phi_{\eta,h} < k(h)$. A deep face of larger size has

$\Phi_{\eta,h} \geq k(h)$, since $\lambda_F \geq 0$. Hence every minimizer is a cycle of size $k(h)$ and depth greater than h, and Theorem 2.5(b) at $\tau = e^{h/\eta}$ puts the support of every global mode among the minimizers for large N. □

Theorem 5.2 is a theorem about least cardinality. It does not say that every cycle of size $k(h)$ and depth greater than h wins, and the cycles that are minimal for inclusion among the deep ones can have other sizes and are then never selected. Among faces of equal size, minimizing $\Phi_{\eta,h}$ is minimizing λ_F^η at fixed η, whatever h is. So the deeper cycle wins first, and at equal depth the smaller prefactor of Proposition 5.3. An exponentially small difference of rates still decides, because $N \to \infty$ first. If two scores agree exactly at some η, the terms of (6) below L may decide, but they need not. In the symmetric case, for instance two disjoint cycles built from the same costs with a start that is symmetric between them, the two faces have the same exit rate and the same constant, and nothing at any order separates them. The theorem then says only that the mode lies on one of the tied cycles, and Section 5.2 treats such ties as unobserved. The full-support start is needed, since otherwise some cycles are not admissible faces.

Proposition 5.3 (prefactors)**.** *For every proper face F there is $a_F \in (0,\infty)$ with $\lambda_F^\eta = a_F e^{-D(F)/\eta}(1+o(1))$, namely $a_F = 1/\rho(B_F)$ with $B_F = \lim_{\eta\downarrow 0} e^{-D(F)/\eta}\mathcal{G}_F(\eta)$, and*

$$D(F) = \nu\big(\det(-Q_F)\big) - \min_{i\in F} \nu\big(\det(-Q_{F\setminus\{i\}})\big),$$

where $\nu(f)$ is the exponent b of the first nonzero term $c\,e^{-b/\eta}$ of a finite sum of exponentials f, and the determinant of the empty matrix is 1.

Proof. Every minor of $-Q_F$ is a finite real combination of exponentials $e^{-b/\eta}$. After equal exponents are collected, a combination that is not identically zero has a first nonzero term $c\,e^{-\nu/\eta}$ and equals $c\,e^{-\nu/\eta}(1+o(1))$. The determinant of $-Q_F$ is positive, since its real eigenvalues are at least $\lambda_F > 0$ and the others come in conjugate pairs. The entries of $\mathcal{G}_F$ are positive by the identity in the proof of Theorem 5.1. By Cramer's rule each is a cofactor divided by $\det(-Q_F)$, so

$$(\mathcal{G}_F)_{ij} = \beta_{ij} e^{e_{ij}/\eta}(1+o(1)), \qquad \beta_{ij} > 0.$$

The diagonal cofactor of j is $\det(-Q_{F\setminus\{j\}})$, so $e_{jj} = \nu(\det(-Q_F)) - \nu(\det(-Q_{F\setminus\{j\}}))$. The exit-time bounds give $\max_j(\mathcal{G}_F)_{jj} \leq 1/\lambda_F \leq |F|\max_j(\mathcal{G}_F)_{jj}$, so $D(F) = \max_j e_{jj}$, which is the determinant formula. Since $(\mathcal{G}_F)_{ij} \leq (\mathcal{G}_F)_{jj}$, we have $e_{ij} \leq D(F)$ for all i,j. Hence $e^{-D(F)/\eta}\mathcal{G}_F$ converges entrywise to the matrix B_F with entries β_{ij} where $e_{ij} = D(F)$ and 0 elsewhere. It has a positive diagonal entry, so $\rho(B_F) > 0$. The spectral radius is continuous in the entries, and $\rho(e^{-D(F)/\eta}\mathcal{G}_F) = e^{-D(F)/\eta}/\lambda_F$. Hence $e^{D(F)/\eta}\lambda_F \to 1/\rho(B_F)$. □

The asymptotics of principal eigenvalues of matrices with entries of order $e^{-V_{ij}/\eta}$, and their expressions through graphs and determinants, are due to Wentzell [28] and Freidlin and Wentzell [15, Chapter 6], and the proof above is a short version for killed generators. The proposition uses the exact form of the rates. A family that satisfies only the logarithmic limits $-\eta\log q_{ij} \to V_{ij}$ need not have finite prefactors.

Theorem 5.4 (critical windows). *Fix a depth $\gamma \in \mathcal{D}$ and $x > 0$, and let $T = xe^{\gamma/\eta}L$. Define the limiting score*

$$\mathcal{J}_{\gamma,x}(F) = \begin{cases} +\infty, & D(F) < \gamma, \\ |F| - 1 + xa_F, & D(F) = \gamma, \\ |F| - 1, & D(F) > \gamma, \end{cases}$$

with V in the last case. If $\mathcal{J}_{\gamma,x}$ has a unique minimizer F_x, then for all sufficiently small η, F_x is the unique minimizer of the score $|F| - 1 + xe^{\gamma/\eta}\lambda_F^\eta$ and the eventual support of the global mode as $N \to \infty$ at that η. The same holds, with one threshold for η, uniformly for x in a compact interval on which one face has a positive gap. The lower envelope $x \mapsto \min_F \mathcal{J}_{\gamma,x}(F)$ is concave and piecewise linear, and its minimizers need not be cycles.

Proof. Let $S_\eta(F) = |F| - 1 + xe^{\gamma/\eta}\lambda_F^\eta$, which is $\Phi_\tau(F)$ for Q_η at $\tau = xe^{\gamma/\eta}$. By Proposition 5.3, $e^{\gamma/\eta}\lambda_F^\eta = a_F e^{(\gamma - D(F))/\eta}(1 + o(1))$, which tends to ∞, a_F or 0 as $D(F) < \gamma$, $D(F) = \gamma$ or $D(F) > \gamma$, and $S_\eta(V) = d - 1$. On a compact interval $I \subset (0, \infty)$ the convergence $S_\eta(F) \to \mathcal{J}_{\gamma,x}(F)$ is uniform in x, divergence included, since x is bounded below on I.

Suppose that one face F satisfies $\mathcal{J}_{\gamma,x}(G) \geq \mathcal{J}_{\gamma,x}(F) + g$ for all $G \neq F$ and $x \in I$, with $g > 0$. There are finitely many faces, so there is η_0 with $S_\eta(G) > S_\eta(F)$ for all $G \neq F$, $x \in I$ and $\eta < \eta_0$. A unique minimizer F_x at a single x has such a gap on a small interval around x, because the scores are continuous in x. Theorem 2.5(b) at $\tau = xe^{\gamma/\eta}$ then gives the support of the global mode, with a threshold in N that may depend on η and x.

The function $x \mapsto \mathcal{J}_{\gamma,x}(F)$ is affine for every face with $D(F) \geq \gamma$, and $\mathcal{J}_{\gamma,x}(V) = d - 1$ is finite. So the envelope is the minimum of finitely many affine functions, which is concave and piecewise linear. Example 5.5 shows a minimizer that is not a cycle. □

At a tie of the limiting lines, Theorem 5.4 identifies the candidates only, and the exact scores at each η decide. Along the envelope the slope a_F of the active line decreases with x, and its intercept $|F| - 1$ increases. The convex hull of first-order winners in [25, Theorem 4.4] behaves in the same way. Putting $h = \gamma$ and treating the faces of depth γ as faces of greater depth would discard the term xa_F, which is of order 1 in the coefficient of L.

Example 5.5 (a noncycle winner on an open window). Take $U = (0, 1, 1, 1)$ and the symmetric saddles $H_{12} = 2$, $H_{13} = H_{23} = H_{34} = 3$, $H_{14} = H_{24} = 5$, so that with $t = e^{-1/\eta}$

$$(q_{ij})_{i \neq j} = \begin{pmatrix} * & t^2 & t^3 & t^5 \\ t & * & t^2 & t^4 \\ t^2 & t^2 & * & t^2 \\ t^4 & t^4 & t^2 & * \end{pmatrix}.$$

The only proper cycle with more than one state is $\{1, 2\}$, of depth 3. The faces of depth 3 are $\{1, 2\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$, with prefactors $a_{12} = 2$, $a_{123} = \frac{2}{3}$ and $a_{124} = 2$, so at $T = xt^{-3}L$ the limiting scores are $1 + 2x$, $2 + \frac{2}{3}x$, $2 + 2x$ and 3 for V. Hence the eventual winner is $\{1, 2\}$ for $0 < x < \frac{3}{4}$, the noncycle $\{1, 2, 3\}$ for $\frac{3}{4} < x < \frac{3}{2}$, and V for $x > \frac{3}{2}$. At $h = 3$ and $x = 1$ the noncycle beats every cycle. Its window has width $\log 2$ in the rescaled height $(h - 3)/\eta$ and collapses onto the single height $h = 3$, which is why it does not contradict Theorem 5.2.

Proof. Here $q_{ij} = t^{H_{ij}-U_i}$ and $\pi \propto (1, t, t, t)$. The exit rates are

$$q_1 = t^2 + t^3 + t^5, \qquad q_2 = t + t^2 + t^4, \qquad q_3 = 3t^2, \qquad q_4 = t^2 + 2t^4.$$

Cycles and depths. The singletons have depths $\min_j V_{ij} = 2, 1, 2, 2$. We test the sets with 2 or 3 states by the energy criterion. The set $\{1, 2\}$ has boundary height 3 and internal saddle $H_{12} = 2 < 3$, so it is a cycle of depth $3 - U_1 = 3$. A set that contains exactly one of the states 1 and 2 has the edge $H_{12} = 2$ to its complement, and no two of its points communicate below 2. The remaining sets $\{3, 4\}$, $\{1, 2, 3\}$ and $\{1, 2, 4\}$ have boundary height 3. In the first two the state 3 is joined to the others only by saddles of height 3, and in the third the state 4 is joined to the others only by saddles of height 5. So none is a cycle. By Theorem 5.1, a face that contains $\{1, 2\}$ has depth 3 and every other proper face has depth at most 2. So $\mathcal{D} = \{1, 2, 3\}$, and the faces of depth 3 are the three named.

Prefactors. We apply Proposition 5.3. For $\{1, 2\}$, $\det(-Q_{12}) = q_1 q_2 - t^3 = 2t^4 + O(t^5)$, and the first column of the adjugate is $(q_2, t) = (t, t) + O(t^2)$, while the second column is $(t^2, q_1) = O(t^2)$. Hence $t^3 \mathcal{G}_{12} \to B_{12}$ with first column $(\frac{1}{2}, \frac{1}{2})$ and second column 0, so $\rho(B_{12}) = \frac{1}{2}$ and $a_{12} = 2$. For $\{1, 2, 3\}$,

$$-Q_{123} = \begin{pmatrix} t^2 + t^3 + t^5 & -t^2 & -t^3 \\ -t & t + t^2 + t^4 & -t^2 \\ -t^2 & -t^2 & 3t^2 \end{pmatrix}.$$

Its cofactors along the first row are $3t^2 q_2 - t^4 = 3t^3 + O(t^4)$, $3t^3 + t^4$ and $t^3 + t^2 q_2 = 2t^3 + O(t^4)$, so expansion along that row gives the determinant $q_1(3t^2 q_2 - t^4) - t^2(3t^3 + t^4) - t^3(t^3 + t^2 q_2) = 2t^6 + O(t^7)$. The cofactors along the second row are $3t^4 + t^5$, $3t^2 q_1 - t^5$ and $t^2 q_1 + t^4$, and along the third row $t^4 + t^3 q_2$, $t^2 q_1 + t^4$ and $q_1 q_2 - t^3$, all $O(t^4)$. So the first column of the adjugate is $(3t^3, 3t^3, 2t^3) + O(t^4)$, its other entries are $O(t^4)$, and B_{123} has first column $(\frac{3}{2}, \frac{3}{2}, 1)$ and zeros elsewhere. Its eigenvalues are $\frac{3}{2}$ and 0, so $a_{123} = \frac{2}{3}$. For $\{1, 2, 4\}$ put $q_4 = t^2 + 2t^4$. The cofactors along the first row are $q_2 q_4 - t^8 = t^3 + O(t^4)$, $t q_4 + t^8 = t^3 + O(t^4)$ and $t^5 + t^4 q_2 = O(t^5)$, so the determinant is $2t^6 + O(t^7)$. The other cofactors are $t^2 q_4 + t^9$, $q_1 q_4 - t^9$, $t^4 q_1 + t^6$, $t^6 + t^5 q_2$, $t^4 q_1 + t^6$ and $q_1 q_2 - t^3$, all $O(t^4)$. So B_{124} has first column $(\frac{1}{2}, \frac{1}{2}, 0)$ and zeros elsewhere, and $a_{124} = 2$.

The winners. At $\gamma = 3$ the faces of depth less than 3 have $\mathcal{J}_{3,x} = \infty$. The lines are $1 + 2x$, $2 + \frac{2}{3}x$, $2 + 2x$ and 3. The face $\{1, 2, 4\}$ is always 1 above $\{1, 2\}$. Then $1 + 2x < 2 + \frac{2}{3}x$ exactly when $x < \frac{3}{4}$, $2 + \frac{2}{3}x < 3$ exactly when $x < \frac{3}{2}$, and $1 + 2x < 3$ exactly when $x < 1$. So the minimizer is unique and equals $\{1, 2\}$ on $(0, \frac{3}{4})$, $\{1, 2, 3\}$ on $(\frac{3}{4}, \frac{3}{2})$ and V on $(\frac{3}{2}, \infty)$, and Theorem 5.4 turns each into the eventual support of the global mode, uniformly on compact subintervals. We claim nothing at the ties $x = \frac{3}{4}$ and $x = \frac{3}{2}$.

The noncycle. At $x = 1$, that is at $h = 3$, the score of $\{1, 2, 3\}$ is $\frac{8}{3}$. The cycles of depth less than 3 have infinite limiting score, and the only cycles of depth at least 3 are $\{1, 2\}$ and V, both of score 3. So the noncycle beats every cycle. In the rescaled height $s = (h - 3)/\eta = \log x$ the window is $\log \frac{3}{4} < s < \log \frac{3}{2}$, of width $\log 2$. Its heights $h = 3 + \eta s$ tend to 3, so at every fixed $h \neq 3$ the window is eventually left. □

The mechanism is that state 3 costs one dimension but returns the chain to $\{1, 2\}$ before it is killed at state 4, which lowers the leading exit rate from $2t^3$ to $\frac{2}{3}t^3$. On the middle

interval this gain is worth more than the dimension. The selected face is the support of the counts, not of the limiting occupation fractions. For $\{1,2,3\}$ the right Perron vector tends to $(1,1,\frac{2}{3})$ up to scale, and with $\pi \propto (1,t,t,t)$ the quasi-ergodic law α^*_{123} tends to the point mass at state 1.

Example 5.6 (a branch switch)**.** Take $U = (2,0,0)$, $H_{12} = H_{13} = 5$ and $H_{23} = 1$. Then $\{2,3\}$ is a cycle of depth 5, the singleton depths are $3,1,1$, and the winners are $\{1\}$ for $0 < h < 3$, $\{2,3\}$ for $3 < h < 5$ and V for $h > 5$. The two winning cycles are disjoint nodes of the same cycle tree, so the selector jumps between branches and never selects the nodes $\{2\}$ and $\{3\}$.

Proof. With $t = e^{-1/\eta}$ the rates are $q_{12} = q_{13} = t^3$, $q_{21} = q_{31} = t^5$ and $q_{23} = q_{32} = t$. So $\lambda_{\{1\}} = 2t^3$, $\lambda_{\{2\}} = \lambda_{\{3\}} = t + t^5$ and $\lambda_{\{2,3\}} = t^5$. By the energy criterion $\{2,3\}$ is a cycle, and $\{1,2\}$, $\{1,3\}$ are not, since each has an edge of height 1 to its complement. Hence $\mathcal{D} = \{1,3,5\}$. Theorem 5.2 gives the winner $\{1\}$ on $(0,3)\setminus\{1\}$, since $2t^3 < t + t^5$, the winner $\{2,3\}$ on $(3,5)$, where $k(h) = 2$, and V above 5. At $h = 1$, $\Phi_{\eta,1}(\{1\}) = 2t^2 \to 0$ while every other face has a score that tends to at least 1, so Theorem 2.5(b) applies directly. We claim nothing at $h = 3$ and $h = 5$. □

5.2. Reading the cycle tree

We observe only the winning face, as a function of the height and of a field, and ask how much of the cycle structure it shows. A field is a vector $v = v(\eta) \in \mathbb{R}^d$, which may depend on η but not on N. As in Section 4.1 we write $\lambda_F(v) = -\varrho(Q_{\eta,F} + \operatorname{diag} v_F)$, with $Q_{\eta,F}$ the killed generator of F under Q_η, and

$$\Phi_{\eta,h}(F;v) = |F| - 1 + e^{h/\eta}\lambda_F(v),$$

so that $\Phi_{\eta,h}(F;0) = \Phi_{\eta,h}(F)$. In this subsection a height h may be any real number. We say that a face G *wins at* (h,v) if there is $\eta_0 > 0$ such that for every $\eta < \eta_0$ the face G is the unique minimizer of $\Phi_{\eta,h}(\cdot\,;v)$ over all nonempty faces. Where two faces of equal size have identical rates, for instance two disjoint cycles built from the same costs, no face wins and nothing is observed.

Proposition 4.1 turns a win into an observation. At a fixed η it applies to Q_η, μ_η, the vector $v(\eta)$ of either sign and $\tau = e^{h/\eta}$. So if G wins at (h,v), then for every $\eta < \eta_0$ the support of every maximizer of the weighted law is G for all large N. Nothing else below is probabilistic.

Two elementary facts are used throughout. The Perron root of a Metzler matrix is nondecreasing in its entries [2, Ch. 2], so $v \le w$ entrywise gives $\lambda_F(w) \le \lambda_F(v)$, and $\lambda_F(v + \kappa\mathbf{1}) = \lambda_F(v) - \kappa$ for every real κ. So adding a constant to the field changes no winner. And by Lemma 2.2, $\lambda_G(v) < \lambda_F(v)$ whenever $F \subsetneq G$. In particular $D(F) \le D(G)$ for $F \subseteq G$.

Proposition 5.7 (the unperturbed path)**.** *Let $v = 0$, and let a proper face W win at some height $h_0 \notin \mathcal{D}$. Then W is a cycle with $D(W) > h_0$, W wins at every $h \in [h_0, D(W)) \setminus \mathcal{D}$, and W wins at no $h > D(W)$. Hence the supremum of the heights at which W wins is $D(W)$.*

Proof. By Theorem 5.2, W is a cycle of size $k(h_0)$ and depth greater than h_0. Let $h \in [h_0, D(W)) \setminus \mathcal{D}$. A face of depth greater than h has depth greater than h_0, so $k(h) \ge$

$k(h_0) = |W|$, and $k(h) = |W|$ since $D(W) > h$. By Theorem 5.2 again, for small η every minimizer of $\Phi_{\eta,h}$ is a cycle G with $|G| = |W|$ and $D(G) > h_0$. For such $G \neq W$, $\Phi_{\eta,h}(G) - \Phi_{\eta,h}(W) = e^{h/\eta}(\lambda_G - \lambda_W)$ has the sign of $\Phi_{\eta,h_0}(G) - \Phi_{\eta,h_0}(W)$, which is positive for small η. Hence W is the unique minimizer. For $h > D(W)$ we have $e^{h/\eta}\lambda_W \to \infty$, while $\Phi_{\eta,h}(V) = d - 1$. □

So the path shows the depth of each face it selects, as the supremum of the heights at which it wins, and nothing about the other faces. The heights at which a face wins may skip depths of faces that never win, and a noncycle winner of a critical window occupies a single height. In Example 5.5 the singleton $\{4\}$ wins on $(0,1) \cup (1,2)$, since $q_4 < q_1 < q_3$ and $q_4 < q_2$ for small η, then $\{1,2\}$ wins on $(2,3)$ and V above 3. The cut at 1 is the depth of the cycle $\{2\}$, which never wins. The path shows $D(\{4\}) = 2$ and $D(\{1,2\}) = 3$ and nothing about the cycles $\{1\}$, $\{2\}$ and $\{3\}$. The reference is absolute, since $\lambda_V(0) = 0$ makes the score of V exactly $d - 1$.

Killing the complement of a face at one fixed large rate does not show its depth either. For two states, $F = \{1\}$ and the killing rate $e^{g/\eta}$ at state 2, the exit rate λ of V is the smaller root of $(q_{12} - \lambda)(q_{21} + e^{g/\eta} - \lambda) = q_{12}q_{21}$, so $\lambda_{\{1\}} - \lambda = q_{12}q_{21}/(q_{21} + e^{g/\eta} - \lambda) = e^{-(V_{12}+V_{21}+g)/\eta}(1 + o(1))$. The winner leaves $\{1\}$ at the height $V_{12} + V_{21} + g$ and not at $D(\{1\}) = V_{12}$. By the shift rule this killing is the same experiment as the reward $e^{g/\eta}$ on $\{1\}$. So the sign of the field is not the issue. Both readings fail for the same reason. They look at the height at which a face stops winning, at one fixed field.

What works is to scan the size of the field on a calibrated scale and to read a threshold in its exponent. For a proper face G, a fixed constant $c > 0$ and a real γ put

$$v^{G,\gamma} = c\, e^{-\gamma/\eta} \mathbf{1}_G.$$

By the shift rule $v^{G,\gamma}$ and the killing field $-c\, e^{-\gamma/\eta}\mathbf{1}_{V\setminus G}$ have the same winners, even at finite N, since the weights differ by the constant factor $e^{cTe^{-\gamma/\eta}}$. We write the field as a reward only because then $v \geq 0$, so $\lambda_V(v) \leq 0$ and $\Phi_{\eta,h}(V; v) \leq d - 1$, which shortens the comparisons. The field is constant on G, so $\lambda_G(v^{G,\gamma}) = \lambda_G - c\, e^{-\gamma/\eta}$ exactly. Define the threshold of G as

$$\vartheta(G) = \sup\big\{\gamma \in \mathbb{R} :\ G \text{ wins at } (h, v^{G,\gamma}) \text{ for some } h > \gamma\big\},$$

with $\sup \emptyset = -\infty$. It is read from the map $(h, v) \mapsto$ winner alone, at heights where a winner exists. The observer needs no list of critical heights, which depend on the depths. The restriction $h > \gamma$ makes the field visible on the time scale, since then $e^{h/\eta} c\, e^{-\gamma/\eta} \to \infty$. Without it every face that wins on the unperturbed path would have an infinite threshold. A strong field, as in the example above with $\gamma = -g$, keeps G winning above its depth, but only the supremum over γ is read.

The next 2 lemmas are standard facts of cycle theory. Laminarity is Catoni's Proposition 4.4 [6] and the tree of cycles of Trouvé [27] and of Olivieri and Vares [24, Chapter 6]. The strict growth of depths along nested cycles can be deduced from Catoni's Propositions 4.5 and 4.10, which express the depth of a cycle through those of its maximal subcycles. We include the short proofs for completeness, in the convention of cycles through hitting probabilities used here.

Lemma 5.8 (laminarity)**.** *Two cycles are disjoint or nested.*

Proof. Let C_1, C_2 be cycles with a common state z, and suppose that neither contains the other. Take $x \in C_1 \setminus C_2$ and $y \in C_2 \setminus C_1$. Since $y \notin C_1$, $\mathbb{P}_z(\tau_x < \tau_y) \geq \mathbb{P}_z(\tau_x < \tau_{C_1^c})$, which tends to 1 because C_1 is a cycle. Since $x \notin C_2$, $\mathbb{P}_z(\tau_{C_2^c} < \tau_y) \geq \mathbb{P}_z(\tau_x < \tau_y)$, which tends to 1 and contradicts that C_2 is a cycle. □

Lemma 5.9 (depth with a calibrated killing)**.** *Let $K \subseteq [d]$, $\gamma \in \mathbb{R}$ and $s = c\, e^{-\gamma/\eta}$. For every proper face G the limit $D^K(G) = \lim_{\eta \downarrow 0}\big(-\eta \log \lambda_G(-s\mathbf{1}_K)\big)$ exists, and*

$$D^K(G) = \max\{D^K(C) : C \subseteq G \text{ a cycle}\}.$$

Moreover $D^K(C) = D(C)$ if $C \cap K = \emptyset$, and $D^K(C) \leq \gamma$ if $C \subseteq K$.

Proof. Fix Λ larger than every element of $\mathcal{D}$. Add a cemetery state ∂. Every state of K jumps to ∂ at rate s, every other state of $[d]$ jumps to ∂ at rate $\varepsilon = e^{-\Lambda/\eta}$, and ∂ jumps to every state of $[d]$ at rate ε. All rates of this enlarged chain are positive. If $\gamma < 0$ we multiply all its rates by $e^{\gamma/\eta}$, which shifts every depth by $-\gamma$ and changes no hitting probability, hence no cycle. The proof of Theorem 5.1 uses the rates only through the limits of $-\eta \log$ of the rates and through the exit-time bounds, once the uniformization constant $\varkappa$ exceeds every exit rate. After the rescaling every rate between two states of $[d]$ is at most 1, every rate into ∂ is at most $\max(c, \varepsilon)$, and ∂ has the exit rate $d\varepsilon$. So every exit rate is at most $d - 1 + \max(c, 1)$, and $\varkappa > d + c$ serves. The bound covers the case where some $V_{ij} = 0$ and $\gamma = 0$, in which the rates 1 and c occur exactly. So the proof applies to the enlarged chain.

For $G \subseteq [d]$ the killed generator of G in the enlarged chain is $Q_G - \operatorname{diag}(s\mathbf{1}_{G\cap K} + \varepsilon \mathbf{1}_{G\setminus K})$. By the two elementary facts its exit rate λ'_G satisfies $\lambda_G(-s\mathbf{1}_K) \leq \lambda'_G \leq \lambda_G(-s\mathbf{1}_K) + \varepsilon$. Since $\lambda_G(-s\mathbf{1}_K) \geq \lambda_G$ and $D(G) < \Lambda$, we get $\lambda'_G = \lambda_G(-s\mathbf{1}_K)(1 + o(1))$. So Theorem 5.1 for the enlarged chain gives the existence of $D^K(G)$ and the identity $D^K(G) = \max D^K(C)$ over the cycles $C \subseteq G$ of the enlarged chain.

Each such C is a cycle of Q_η. Indeed, the enlarged chain is the chain X of Q_η run until an independent killing time, and then ∂. If X leaves C before it hits y, so does the enlarged chain. So for distinct $x, y \in C$ the probability $\mathbb{P}_x(\tau_{C^c} < \tau_y)$ for X is at most the same probability for the enlarged chain, which is exponentially small. Hence $D^K(G)$ is at most the maximum of $D^K(C)$ over the cycles $C \subseteq G$ of Q_η, and the reverse inequality holds by Lemma 2.2. If $C \cap K = \emptyset$ then $\lambda_C(-s\mathbf{1}_K) = \lambda_C$. If $C \subseteq K$ then $\lambda_C(-s\mathbf{1}_K) = \lambda_C + s \geq s$, so $D^K(C) \leq \gamma$. □

Lemma 5.10 (strict growth along nested cycles)**.** *If $C' \subsetneq C$ are cycles, then $D(C') < D(C)$. Hence $D(S) < D(C)$ for every nonempty proper subset S of a cycle C.*

Proof. Fix $z \in C \setminus C'$. Since C is a cycle there is $\alpha > 0$ with $\mathbb{P}_w(\tau_{C^c} < \tau_y) \leq e^{-\alpha/\eta}$ for all distinct $w, y \in C$ and all small η. For each η choose $y \in C'$ with $\mathbb{E}_y \tau_{C'^c} = M^{\text{exit}}_{C'}$. Start at y and cut the path into rounds. A round runs until the chain leaves C'. If it then sits at a state $w \in C \setminus C'$, the round continues until the chain returns to y or leaves C. A round ends outside C only if the chain leaves C' straight into C^c, which forces $\tau_{C^c} < \tau_z$, or if it leaves C from w before it reaches y. By the strong Markov property each round ends outside C with probability at most $2e^{-\alpha/\eta}$, and the rounds are independent and identically distributed. So the number R of rounds up to the exit from C has $\mathbb{E}R \geq e^{\alpha/\eta}/2$. Each round lasts at least its first part, of mean $M^{\text{exit}}_{C'}$, and R is a stopping time for the rounds. Wald's identity gives

$$M^{\text{exit}}_C \geq \mathbb{E}_y \tau_{C^c} \geq \mathbb{E}R \cdot M^{\text{exit}}_{C'} \geq \tfrac{1}{2} e^{\alpha/\eta} M^{\text{exit}}_{C'},$$

and $D(C) \geq D(C') + \alpha$ by the last statement of Theorem 5.1. For the second claim, Theorem 5.1 gives a cycle $C'' \subseteq S$ with $D(C'') = D(S)$, and $C'' \subsetneq C$. □

Theorem 5.11 (depths from a calibrated attraction)**.** *Fix $c > 0$.*

(a) Let G be a proper face. If G wins at $(h, v^{G,\gamma})$ for some $h > \gamma$, then $\gamma \leq D(G)$. Hence $\vartheta(G) \leq D(G)$.

(b) Let C be a cycle, and put $D^-(C) = \max_{i \in C} D(C \setminus \{i\})$, with $D^-(C) = -\infty$ for a singleton. Then $D^-(C) < D(C)$, and for every $\gamma < D(C)$ the cycle C wins at $(h, v^{C,\gamma})$ for every $h \in (\max(\gamma, D^-(C)), D(C))$. Hence $\vartheta(C) = D(C)$.

(c) For every proper face F,

$$D(F) = \max_{G \subseteq F} \vartheta(G).$$

Proof. Write $s = c\, e^{-\gamma/\eta}$ and $\sigma = e^{h/\eta}$, and $\Phi(G)$ for the score of G under the field.

(a) Let $\gamma > D(G)$ and $h > \gamma$. Then s is of smaller order than λ_G, so $\sigma(\lambda_G - s) \geq \sigma\lambda_G/2 \to \infty$. Since $v^{G,\gamma} \geq 0$, the score of V is at most $d - 1$, so G is not a minimizer for small η.

(b) The inequality $D^-(C) < D(C)$ is Lemma 5.10. Fix $h \in (\max(\gamma, D^-(C)), D(C))$ and write $v = v^{C,\gamma}$. The score of C is $|C| - 1 + \sigma\lambda_C - \sigma s$, where $\sigma\lambda_C \to 0$ because $h < D(C)$ and $\sigma s = c\, e^{(h-\gamma)/\eta} \to \infty$ because $h > \gamma$. On another face G the field is $s\mathbf{1}_{G \cap C} = s\mathbf{1}_G - s\mathbf{1}_{G \setminus C}$, so

$$\lambda_G(v) = \lambda_G(-s\mathbf{1}_{G \setminus C}) - s.$$

If $G \cap C = \emptyset$, then $\lambda_G(v) = \lambda_G > 0$, so $\Phi(G) \geq 0$ while $\Phi(C) \to -\infty$. *If $G \subsetneq C$*, then $\lambda_G(v) - \lambda_C(v) = \lambda_G - \lambda_C$, and $D(G) \leq D^-(C) < h < D(C)$, so $\sigma(\lambda_G - \lambda_C) \geq \sigma\lambda_G/2 \to \infty$. *If $G \supsetneq C$*, then $\lambda_C(v) - \lambda_G(v) = \lambda_C - \lambda_G(-s\mathbf{1}_{G \setminus C}) \leq \lambda_C$, so $\Phi(G) - \Phi(C) \geq |G| - |C| - \sigma\lambda_C \to |G| - |C| \geq 1$. The face V is of this kind. *Otherwise* G meets C without containing it or lying in it. Put $K = G \setminus C$. Then $\lambda_G(v) - \lambda_C(v) = \lambda_G(-s\mathbf{1}_K) - \lambda_C$. A cycle $C' \subseteq G$ that met both C and K would be nested with C by Lemma 5.8, which is impossible since C' meets K and $C \not\subseteq G$. So C' lies in $G \cap C$ or in K. In the first case Lemma 5.9 gives $D^K(C') = D(C') \leq D(G \cap C) \leq D^-(C)$, and in the second $D^K(C') \leq \gamma$. By Lemma 5.9, $D^K(G) \leq \max(D^-(C), \gamma) < h$, so $\sigma\lambda_G(-s\mathbf{1}_K) \to \infty$, while $\sigma\lambda_C \to 0$.

In every case $\Phi(G) - \Phi(C)$ tends to $+\infty$ or has limit inferior at least 1. There are finitely many faces, so C wins at (h, v). The interval of h is nonempty for every $\gamma < D(C)$, so $\vartheta(C) \geq D(C)$, and (a) gives equality.

(c) For $G \subseteq F$, (a) and the monotonicity of D give $\vartheta(G) \leq D(G) \leq D(F)$. By Theorem 5.1 there is a cycle $C \subseteq F$ with $D(C) = D(F)$, and $\vartheta(C) = D(F)$ by (b). □

The theorem needs no uniqueness. Under its own attraction the score of C tends to $-\infty$, so C wins even when other faces tie, as for two disjoint cycles of equal rates. The unperturbed path has no such property. Only the threshold in γ is used, since above $D(C)$ the faces that contain C compete through $\sigma(\lambda_C(v) - \lambda_G(v))$, which can diverge. A noncycle may win under its own attraction and its threshold may reach $D(G)$, which is why (c) takes the maximum over all subfaces and needs no prior knowledge of the cycles. In the deep-singleton family with $V_{12} = V_{21} = 1$, $V_{13} = V_{23} = 3$ and $V_{31} = V_{32} = 5$, the singleton $\{3\}$ of depth 5 wins on all of $(0, 5)$ along the unperturbed path, so the path never shows the cycle $\{1, 2\}$ of depth 3, while Theorem 5.11(b) reads $\vartheta(\{1, 2\}) = 3$.

Theorem 5.12 (cycles from depths)**.** *For a nonempty proper face C the following are equivalent.*

(i) C is a cycle.

(ii) $D(C') < D(C)$ for every nonempty proper subset C' of C.

(iii) $|C| = 1$, or $D(C \setminus \{i\}) < D(C)$ for every $i \in C$.

Moreover $\Gamma(C) = D(C)$ for every cycle. Hence the depth function determines the set of cycles, their depths and the cycle tree, that is, the laminar family of cycles ordered by inclusion with root V, and depths increase strictly along every branch.

Proof. (i) implies (ii) by Lemma 5.10, and (ii) and (iii) are equivalent since D is monotone. Suppose (ii). Theorem 5.1 gives a cycle $C^* \subseteq C$ with $D(C^*) = D(C)$, and (ii) forces $C^* = C$. The identity $\Gamma(C) = D(C)$ is in Theorem 5.1, laminarity is Lemma 5.8, and the strict growth is Lemma 5.10. □

No nondegeneracy is assumed. Ties between the depths of different faces are allowed, because the cycles are strict and Lemma 5.10 rules out a tie along nested cycles. The weak form of the criterion fails. Take 3 states with all $V_{ij} = 1$. Then $\lambda_{\{1\}} = 2e^{-1/\eta}$ and $\lambda_{\{1,2\}} = e^{-1/\eta}$, so $D(\{1\}) = D(\{2\}) = D(\{1,2\}) = 1$. The set $\{1,2\}$ is not a cycle, since from state 1 the chain leaves it before it hits 2 with probability $\frac{1}{2}$. Criterion (ii) rejects it, and the weak criterion $D(C') \le D(C)$ accepts it. In Example 5.5 the faces $\{1,2,3\}$ and $\{1,2,4\}$ tie in depth with the cycle $\{1,2\}$ and are rejected for the same reason.

Corollary 5.13 (the cycle tree from the functionals)**.** *Let (Q_η) and (Q'_η) be Arrhenius families on $[d]$ with all rates positive, and suppose that $J_V^\eta = J_V'^\eta$ on the simplex for every η in a sequence that tends to 0. Then $\lambda_F^\eta = \lambda_F'^\eta$ for every face F and every such η. So the two families have the same depths, the same cycles, the same cycle tree and the same prefactors. The same holds with the first-order modal-support maps of Q_η and Q'_η, for a start of full support, in place of J_V^η.*

Proof. At each η both generators are irreducible, since all rates are positive. Theorem 3.3 gives $\lambda_F^\eta = \lambda_F'^\eta$ for every face, and Theorem 4.6 gives the same conclusion from equal modal-support maps. The depths and the prefactors are limits as $\eta \downarrow 0$, which exist by Theorem 5.1 and Proposition 5.3, so they may be taken along the sequence. The cycles and the tree follow by Theorem 5.12. □

The depths are even explicit in the principal minors, by the determinant formula of Proposition 5.3. Corollary 5.13 uses the whole functional, or the whole map, at each temperature. In the case of the maps the descent of Section 4.2 chooses its fields after Q_η is known. Theorem 5.11 uses one family of fields, fixed in advance, and reads the depths as thresholds. Theorems 5.11 and 5.12 and Corollary 5.13 together give Theorem C.

The prefactors are read at a critical depth from the unperturbed path. Fix $\gamma \in \mathcal{D}$, take $v = 0$ and $T = xe^{\gamma/\eta}L$, and let $\mathcal{J}_{\gamma,x}$ be the limiting score of Theorem 5.4. This is the height $h = \gamma + \eta \log x$, which moves with η, so here we observe the winner as a function of the window coordinate x. Write $e(x) = \min_F \mathcal{J}_{\gamma,x}(F)$ for the lower envelope, and put $\bar{a}_G = a_G$ if $D(G) = \gamma$ and $\bar{a}_G = 0$ if $D(G) > \gamma$, so that $\mathcal{J}_{\gamma,x}(G) = |G| - 1 + x\bar{a}_G$ when $D(G) \ge \gamma$.

Theorem 5.14 (prefactors from window endpoints). *Suppose there are* $0 = x_0 < x_1 < \cdots < x_m < x_{m+1} = \infty$ *and faces* $G_0, \dots, G_m$ *with* $G_{k-1} \neq G_k$ *such that* G_k *is the unique minimizer of* $\mathcal{J}_{\gamma,x}$ *for every* $x \in (x_k, x_{k+1})$.

(a) $D(G_m) > \gamma$, *and* $D(G_k) = \gamma$ *for* $k < m$.

(b) $\bar{a}_{G_m} = 0$, *and for* $k = m, \dots, 1$,

$$\bar{a}_{G_{k-1}} = \bar{a}_{G_k} + \frac{|G_k| - |G_{k-1}|}{x_k}.$$

Hence the points x_k *and the sizes determine* a_{G_k} *for every* $k < m$.

(c) Let G *be a face of depth* γ *other than* $G_0, \dots, G_m$, *and put* $s_G = \sup_{x>0}(e(x) - |G| + 1)/x$. *Then* $a_G \geq s_G$, *and every value* $a > s_G$ *in place of* a_G *gives the same unique minimizers. If* $G' \subseteq G$ *has depth* γ, *then* $a_G \leq a_{G'}$.

Proof. By Theorem 5.4, G_k is the eventual winner, uniformly on compact subintervals of (x_k, x_{k+1}), so the faces G_k and the points x_k are observed. The envelope is continuous, so the lines of G_{k-1} and G_k meet at x_k, that is, $|G_{k-1}| - 1 + x_k \bar{a}_{G_{k-1}} = |G_k| - 1 + x_k \bar{a}_{G_k}$. This is the recursion in (b). Since G_{k-1} lies below G_k to the left of x_k and above it to the right, $\bar{a}_{G_{k-1}} > \bar{a}_{G_k} \geq 0$, and a face with $\bar{a}_G > 0$ has depth γ. As $x \to \infty$ every line of positive slope diverges, while $\mathcal{J}_{\gamma,x}(V) = d - 1$. So the last piece has slope 0, that is, $D(G_m) > \gamma$ and $\bar{a}_{G_m} = 0$. This proves (a) and (b). For (c), the envelope is the minimum of the lines, so $|G| - 1 + x a_G \geq e(x)$ for all $x > 0$, which is $a_G \geq s_G$. If $a > s_G$, then $|G| - 1 + xa > e(x)$ for all $x > 0$, and the unique minimizers do not change. Finally $\lambda_G \leq \lambda_{G'}$ for $G' \subseteq G$, and at equal depths Proposition 5.3 turns this into $a_G \leq a_{G'}$. □

The reference is absolute because the last piece has slope exactly 0, and V, with $\lambda_V(0) = 0$, guarantees a face with a flat line. In Example 5.5 at $\gamma = 3$ the pieces are $G_0 = \{1, 2\}$, $G_1 = \{1, 2, 3\}$ and $G_2 = V$, with $x_1 = 3/4$, $x_2 = 3/2$ and sizes $2, 3, 4$. The recursion gives $a_{123} = (4-3)/(3/2) = 2/3$ and $a_{12} = 2/3 + (3-2)/(3/4) = 2$, the values found in the example. The face $\{1, 2, 4\}$ is never on the envelope. Here $s_{124} = 2/3$, attained on the whole interval $(3/4, 3/2)$, and $\{1, 2\} \subseteq \{1, 2, 4\}$ gives $a_{124} \leq a_{12} = 2$. So $2/3 \leq a_{124} \leq 2$. At $a_{124} = 2/3$ the line of $\{1, 2, 4\}$ would coincide with that of $\{1, 2, 3\}$ on the middle interval, so the hypothesis of Theorem 5.14 excludes the lower end, but the observed winners alone do not. The window thus recovers $a_{12} = 2$ and $a_{123} = 2/3$ and only bounds a_{124}, whose value is 2. A calibrated field need not help, because a field changes the rate of V to a value that depends on Q. In this example the limiting quasi-stationary laws of the 3 faces of depth 3 and the limit of π all sit at state 1. So a field $c\, e^{-3/\eta}$ at state 1 moves every line of the window by $-cx$ to first order, and a field of this order at another state moves no line.

Remark 5.15 (what the design needs). The design of Theorem 5.11 needs the temperature η, the time unit, the amplitude c and the labels of the states, and no rate of Q_η. It observes the winner along a family $\eta \downarrow 0$, with $N \to \infty$ first at each η. No step of this subsection uses reversibility. Theorem 5.1 rests on Catoni's theorem and the exit-time bounds, and the other lemmas use only the Green matrix, the definition of a cycle and the strong Markov property. Lemma 5.9 applies Theorem 5.1 to a chain with a cemetery, which is never reversible. The selection step is Proposition 4.1, which holds for a general generator.

6. Computations and open questions

Every statement is proved in the text, with one kind of exception. The explicit numbers in the example after Corollary 3.6 were computed, in rational arithmetic for the example, and the scripts below check them. The scripts also check other parts of the paper by assertions, with a tolerance in floating point.

The script `verify_reconstruction.py` checks Sections 3 and 4. On 64 random generators with a rank-one cut it checks that the generator point of the cut-transpose has the principal minors of $-Q$ (in 49 of them it is neither Q nor Q^*), and it checks the four-state example after Corollary 3.6 in rational arithmetic. It checks the derivative used in the proof of Lemma 3.2, the three-state identities of Remark 3.7 symbolically, and on 200 generators with 3 states that every generator with the same minors is Q or Q^*. The script `verify_cycles.py` redoes Examples 5.5 and 5.6 in exact arithmetic. The script `verify_attraction.py`, with the module `attraction_core.py`, tests Theorem 5.11 on 11 cost matrices and 138 proper faces at $t = 10^{-32}$ and 10^{-48}, with the exact finite-η scores. Every cycle has the grid threshold $D(G) - 1/4$, no noncycle exceeds it, and $\max_{G \subseteq F} \vartheta(G)$ gives $D(F)$ on every face. The repository also holds the scripts for the reversible design of the arXiv version.

Some finite and algebraic steps are also proved in Lean 4 with Mathlib, in `lean/PaperC` of the same repository. The module `PrincipalMinors` proves the multiaffine expansion of R_F. The module `GeneratorFiber` proves the normalization (N), the fact that a diagonal similarity keeping an irreducible Metzler matrix Metzler has entries of one sign, and Corollary 3.6(d) for positive rates. The module `CriticalWindow` proves the determinants, adjugates and prefactors of Example 5.5, and `DVRestriction` the restriction identity of Proposition 2.3(a). Mathlib has no Perron and Frobenius theorem for irreducible Metzler matrices, so the positive null vector in (N) is a hypothesis there. The library builds with `lake build PaperC`, and every theorem in it depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The build log is `lean/build-log.txt`.

Open questions. The finite-horizon label $\mathcal{W}_N$ of Section 4.1 may determine more than W_Q. The crossing equation of [25, Theorem 5.8] for the weighted law would read the amplitudes of Proposition 3.9 off the positions of the changes of $\mathcal{W}_N$, and so separate the fiber of Corollary 3.6 by finite-horizon labels, but this is not proved. We do not know whether a single irreducible face is always traced against a singleton rather than a larger subface in the exposure lemma. At low temperature it is open whether some calibrated field separates a_{124} from a_{12} in Example 5.5, and a version of Theorem 5.11 with η tied to N is open.

Data and code availability

The code and data that support this paper are in the folder `kagey-problems/problem131/paperC2` of https://github.com/apemm/Kagey-Problems. The command `py -I verify_all.py`, run there with Python 3, `numpy`, `scipy`, `sympy` and `mpmath`, reruns the checks of Section 6. Each output goes to `paperC2/data`, headed by the sha256 of its script.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work grew out of Problem 131 of Peter Kagey's Open Problem Collection [19]. We thank him for posing it.

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